Does high-velocity resistance exercise elicit greater physical function benefits than traditional resistance exercise in older adults? A systematic review and network meta-analysis of 79 trials

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eAppendix 1. Search strategy

Search strategy for PubMed search

(Old [title/abstract] OR Older [title/abstract] OR Elderly [title/abstract] OR Senior [title/abstract] OR Geriatric [title/abstract]) AND ("Power Training" [title/abstract] OR "High-Speed Resistance Training" [title/abstract] OR "High-Velocity Resistance Training" [title/abstract] OR "High-Velocity Strength Training" [title/abstract] OR "Explosive-Type Resistance Training" [title/abstract] OR "Explosive-Type Strength Training" [title/abstract] OR "Resistance Exercise" [title/abstract] OR "Resistance Training" [title/abstract] OR Training, Resistance [title/abstract] OR "Strength Exercise" OR "Strength Training" [title/abstract] OR Training, Strength [title/abstract] OR Weight-Lifting Strengthening Program [title/abstract] OR Strengthening Program, Weight-Lifting [title/abstract] OR Strengthening Programs, Weight-Lifting [title/abstract] OR Weight Lifting Strengthening Program [title/abstract] OR Weight-Lifting Strengthening Programs [title/abstract] OR Weight-Lifting Exercise Program [title/abstract] OR Exercise Program, Weight-Lifting [title/abstract] OR Exercise Programs, Weight-Lifting [title/abstract] OR Weight Lifting Exercise Program [title/abstract] OR Weight-Lifting Exercise Programs [title/abstract]) AND ("Walking Speed" [title/abstract] OR "Gait Speed" [title/abstract] OR "Chair Rise" [title/abstract] OR "Chair Rising" [title/abstract] OR "Chair Stand" [title/abstract] OR "Sit To Stand" [title/abstract] OR "Timed Up And Go" [title/abstract] OR "Timed Get Up And Go" [title/abstract] OR "Up and Go" [title/abstract] OR "Up & Go" [title/abstract] OR TUG [title/abstract] OR "Six Minute Walk Test" [title/abstract] OR "6 Minute Walk Test" [title/abstract] OR "6 Min Walk Test" [title/abstract] OR Power [title/abstract] OR "Leg Press" [title/abstract]) AND ("Clinical Trial" (Mesh) OR "Randomized Controlled Trial" [title/abstract] OR "Controlled Clinical Trial" [title/abstract] OR "Randomized" [title/abstract] OR "Randomly" [title/abstract] OR Trial [title/abstract] OR Control [title/abstract]) NOT (Review [title/abstract] OR "Systematic Review" [title/abstract] OR Meta-analysis [title/abstract] OR "Cross-Sectional study" [title/abstract] OR Protocol [title/abstract])

Search strategy for CINAHL, SportDiscus and Web of Science search

(Old OR Older OR Elderly OR Senior OR Geriatric) AND ("Power Training" OR "High-Speed Resistance Training" OR "High-Velocity Resistance Training" OR "High-Velocity Strength Training" OR "Explosive-Type Resistance Training" OR "Explosive-Type Strength Training" OR "Resistance Exercise" OR "Resistance Training" OR Training, Resistance OR "Strength Exercise" OR "Strength Training" OR Training, Strength OR Weight-Lifting Strengthening Program OR Strengthening Program, Weight-Lifting OR Strengthening Programs, Weight-Lifting OR Weight Lifting Strengthening Program OR Weight-Lifting Strengthening Programs OR Weight-Lifting Exercise Program OR Exercise Program, Weight-Lifting OR Exercise Programs, Weight-Lifting OR Weight Lifting Exercise Program OR Weight-Lifting Exercise Programs) AND ("Walking Speed" OR "Gait Speed" OR "Chair Rise" OR "Chair Rising" OR "Chair Stand" OR "Sit To Stand" OR "Timed Up And Go" OR "Timed Get Up And Go" OR TUG OR "Up and Go" OR "Up & Go" OR "Six Minute Walk Test" OR "6 Minute Walk Test" OR Power OR "Leg Press") AND ("Clinical Trial" OR "Randomized Controlled Trial" OR "Controlled Clinical Trial" OR "Randomized" OR "Randomly" OR Trial OR Control) NOT (Review OR "Systematic Review" OR Meta-analysis OR "Cross-Sectional study" OR Protocol)

Search strategy for EMBASE search

('older adult' OR aged OR geriatric) AND ('resistance training' OR 'power training') AND ('physical performance' OR 'physical function' OR 'walking speed' OR 'chair rise test' OR 'chair rising test' OR 'chair stand' OR 'chair stand test' OR 'sit to stand' OR 'sit to stand movement' OR 'timed up and go' OR 'timed get up and go' OR 'timed up and go test' OR ' Six Minute Walk Test' OR power OR 'leg press') AND ('clinical trial' OR 'randomized controlled trial' OR 'controlled study')

Search strategy for LILACS and Scielo

(idoso OR mayor OR anciano OR viejo) AND (treinamento de força OR treinamento resistido OR musculação OR entrenamiento de resistencia OR entrenamiento de fuerza)

eTable 1. Study characteristics: experimental design and sample size, exercise prescription and outcomes assessed.

Author, year	Participants' characteristics	Experimental design	Exercise intervention prescription	Adverse events	Outcomes
Charette et al., 1991	19 physically healthy older women Age: mean of 69.1 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 13, 3 sessions per week for 12 weeks performing a supervised program undertaking 3-6 sets at 65-75% of 1-RM in 7 exercises. Participants were instructed to perform concentric and eccentric phases in 2 and 3 sec, respectively.	NR	Leg press 1-RM
Judge et al., 1994	55 physically healthy older men and women Age: mean of 80.5 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 28, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 12 reps at 60-75% of 1-RM in 6 exercises.	NR	Fast walking speed;
McCartney et al., 1995	142 physically healthy older men and women Age: mean of 68.2 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 76, 2 sessions per week for 42 weeks performing a supervised program undertaking 2-3 sets of 10-12 reps at 50-80% of 1-RM in 4 exercises.	No adverse events related to study intervention.	Leg press 1-RM
Sipilä et al., 1996	23 physically healthy older women Age: ranging from 76 to 78 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 12, 3 sessions per week for 18 weeks performing a supervised program undertaking 3-4 sets of 8-10 reps at 60-75% of 1-RM in 4 exercises. Participants were instructed to perform both concentric and eccentric phases within 4-6 sec, respectively.	NR	Fast walking speed
Taaffe et al., 1999	46 physically healthy older men and women Age: ranging from 65 to 79 yrs BMI: mean of 24.4 $kg.m^{-2}$	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 34, 1, 2 or 3 sessions per week for 24 weeks performing a supervised program undertaking 3 sets of 8 reps at 60-80% of 1-RM in 8 exercises.	NR	Leg press 1-RM
Haykowsky et al., 2000	22 physically healthy older men	Traditional resistance exercise vs.	**Traditional resistance exercise**	No adverse events related to study intervention.	Leg press 1-RM

	Age: mean of 68 yrs BMI: NR	Non-active control group	n= 12, 3 sessions per week for 16 weeks performing a supervised program undertaking 3 sets of 10 reps at 60-80% of 1-RM in 8 exercises.		
Rhodes et al., 2000	44 physically healthy older women Age: mean of 68.8 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 22, 3 sessions per week for 52 weeks performing a supervised program undertaking 3 sets of 8 reps at 75% of 1-RM in 6 exercises. Participants were instructed to perform concentric and eccentric phases in 2-3 and 3-4 sec, respectively.	NR	Leg press 1-RM
Fatouros et al., 2002	16 physically healthy older men Age: mean of 70.4 yrs BMI: mean of 29.3 $kg.m^{-2}$	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 8, 3 sessions per week for 16 weeks performing a supervised program undertaking 2-3 sets of 8-14 reps at 55-80% of 1-RM in 8 exercises. Participants were instructed to perform concentric and eccentric phases in 2 and 2-3 sec, respectively.	NR	Leg press 1-RM
Fielding et al., 2002	30 physically impaired older women (self-reported disability) Age: mean of 73.0 yrs BMI: mean of 30.1 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 15, 3 sessions per week for 16 weeks performing a supervised program undertaking 3 sets of 8 reps at 70% of 1-RM in 2 exercises. Participants were instructed to perform each repetition as fast as they could. **Traditional resistance exercise** n= 15, 3 sessions per week for 16 weeks performing a supervised program undertaking 3 sets of 8 reps at 70% of 1-RM in 2 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	Two participants in each group reported an exacerbation of preexisting osteoarthritis, while one participant in the high-velocity resistance exercise group reported a recurrence of chronic plantar fasciitis.	Leg press muscle power
Miszko et al., 2003	39 physically healthy older men and women Age: mean of 72.5 yrs	High-velocity resistance exercise vs.	**High-velocity resistance exercise** n= 11, 3 sessions per week for 16 weeks performing a supervised	No adverse events related to study intervention.	Leg press 1-RM

	BMI: NR	Traditional resistance exercise vs. Educational control group	program undertaking 3 sets of 6-8 reps at 40% of 1-RM in 8 exercises. Participants were instructed to perform concentric and eccentric phases in ~1 and ~2 sec, respectively. **Traditional resistance exercise** n= 13, 3 sessions per week for 16 weeks performing a supervised program undertaking 3 sets of 6-8 reps at 50-70% of 1-RM in 8 exercises. Participants were instructed to perform concentric phase in ~4 sec.		
Bean et al., 2004	21 physically impaired older women (self-reported disability) Age: mean of 78.0 yrs BMI: mean of 26.3 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 11, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 10 reps in 6 exercises. Participants were instructed to perform each repetition as fast as they could. **Traditional resistance exercise** n= 13, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 10 reps in 11 exercises. Participants were instructed to perform concentric phase in 2 sec.	No adverse events related to study intervention.	Chair rise test; Leg press muscle power
Boshuizen et al., 2005	33 physically frail older men and women Age: mean of 78.6 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 16, 3 sessions per week for 10 weeks performing a supervised program undertaking 3 sets of 4-8 RMs in 9 exercises. Participants were instructed to perform concentric phase in 2 sec.	NR	Timed-up and go test
de Vos et al., 2005	112 physically healthy older men and women Age: mean of 69 yrs BMI: mean of 26.0 $kg.m^{-2}$	High-velocity resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 84, 2 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8 reps at 20, 50 or 80% of 1-RM in 5 exercises. Participants were instructed to perform	Two participants reported minor strains, one participant reported tendonitis, and one participant reported	Leg press muscle power

			the concentric phase as fast as they could, and the eccentric phase over 3 sec.	exacerbation of osteoarthritis.	
de Vreede et al., 2005	65 physically healthy older women Age: mean of 73.9 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 34, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 10 reps in 15 exercises.	One participant reported a strained hamstring muscle during the training period.	Timed-up and go test; Leg press muscle power
Sousa & Sampaio, 2005	20 physically healthy older men Age: mean of 74.0 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 10, 3 sessions per week for 14 weeks performing a supervised program undertaking 2-3 sets of 6-12 reps at 50-80% of 1-RM in 7 exercises.	NR	Timed-up and go test
Henwood & Taaffe, 2006	67 physically healthy older men and women Age: mean of 70.0 yrs BMI: mean of 26.6 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 23, 2 sessions per week for 8 weeks performing a supervised program undertaking 3 sets of 8 reps at 45-75% of 1-RM in 5-6 exercises. Participants were instructed to perform the concentric phase as fast as they could and the eccentric phase in 3 sec. **Traditional resistance exercise** n= 22, 2 sessions per week for 8 weeks performing a supervised program undertaking 3 sets of 8 reps at 75% of 1-RM in 5-6 exercises. Participants were instructed to perform both concentric and eccentric phases in 3 sec.	NR	Chair rise test, Leg press 1-RM Stair climbing muscle power
Bottaro et al., 2007	20 physically healthy older men Age: mean of 66.5 yrs BMI: mean of 21.2 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 11, 2 sessions per week for 10 weeks performing a supervised program undertaking 3 sets of 8-10 reps at 40-60% of 1-RM in 7 exercises. Participants were instructed to perform concentric and eccentric phases in ~1 and 2-3 sec, respectively. **Traditional resistance exercise**	NR	Timed-up and go test; 30-sec chair rise test; Leg press 1-RM; Leg press muscle power

			n= 9, 2 sessions per week for 10 weeks performing a supervised program undertaking 3 sets of 8-10 reps at 40-60% of 1-RM in 7 exercises. Participants were instructed to perform both concentric and eccentric phases in 2-3 sec.		
Fahlman et al., 2007	72 physically impaired older men and women (functionally limited) Age: mean of 75.5 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 39, 2 sessions per week for 10 weeks performing a supervised program undertaking 1-2 sets of 10-12 reps in 13 exercises.	NR	30-sec chair rise test; 6-min walking test
Caserotti et al., 2008	40 physically healthy and 25 physically impaired older women (accelerated physical impairments) Age: mean of 70.0 yrs BMI: NR	High-velocity resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 32, 2 sessions per week for 12 weeks performing a supervised program undertaking 4 sets of 8-10 reps at 75-80% of 1-RM in 5 exercises. Participants were instructed to perform the concentric phase as fast as they could, and the eccentric phase in a moderate-to-slow speed.	One participant reported an exacerbation of preexisting osteoarthritis.	Countermovement jump muscle power; Leg press muscle power
Reid et al., 2008	57 physically impaired older men and women (mobility limitations) Age: mean of 74.2 yrs BMI: mean of 29.0 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise vs. Stretching control group	**High-velocity resistance exercise** n= 23, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8 reps at 70% of 1-RM in 2 exercises. Participants were instructed to perform the concentric phase as fast as they could, and the eccentric phase over 2 sec. **Traditional resistance exercise** n= 22, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8 reps at 70% of 1-RM in 2 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	No adverse events related to study intervention.	Leg press muscle power

Bean et al., 2009	138 physically impaired older men and women (mobility limitations) Age: mean of 75.4 yrs BMI: NR	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 72, 3 sessions per week for 16 weeks performing a supervised program undertaking 2 sets in 10 exercises. Participants were instructed to perform the concentric phase as fast as they could. **Traditional resistance exercise** n= 66, 3 sessions per week for 16 weeks performing a supervised program undertaking 2 sets of 10 reps in 11 exercises. Participants were instructed to perform both concentric and eccentric phases in 3 sec.	Two participants reported muscle pain resulting from the exercise programs and dropped out.	Leg press muscle power
Marsh et al., 2009	45 physically impaired older men and women (mild-to-moderate self-reported disability) Age: mean of 74.8 yrs BMI: mean of 30.4 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 15, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8-10 reps at 70% of 1-RM in 7 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 2-3 sec. **Traditional resistance exercise** n= 15, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8-10 reps at 70% of 1-RM in 7 exercises. Participants were instructed to perform both concentric and eccentric phases in 2-3 sec.	No adverse events related to study intervention.	Leg press muscle power
Kalapotharakos et al., 2010	14 physically healthy older men Age: mean of 83.0 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 7, 2 sessions per week for 14 weeks performing a supervised program undertaking 3 sets at 70% of 3-RM in 6 exercises.	NR	Timed-up and go test; Chair rise test; 6-min walking test
Lustosa et al., 2011	48 physically frail older women; Age: mean of 72.0 yrs	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 32, 3 sessions per week for 10 weeks.	NR	Timed-up and go test

	BMI: mean of 29.1 kg.m^{-2}				
Correa et al., 2012 & Pinto et al., 2014	58 physically healthy older women Age: mean of 67.0 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 41, 2 session per week for 6 weeks performing a supervised program undertaking 2-3 sets of 12-20 RMs in 3 exercises.	NR	Timed-up and go test; 30-sec chair rise test
Pereira et al., 2012	56 physically healthy older women Age: mean of 62.5 yrs BMI: mean of 27.6 kg.m^{-2}	High-velocity resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 28, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 4-10 reps at 40-75% of 1-RM in 6 exercises. Participants were instructed to perform the concentric phase as fast as they could.	NR	Timed-up and go test; 30-sec chair rise test
Raj et al., 2012	22 physically healthy older women Age: mean of 67.5 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 12, 2 session per week for 16 weeks performing a supervised program undertaking 2 sets of 10 reps at 75% of 1-RM in 4 exercises.	NR	Timed-up and go test; Stair climb muscle power; Countermovement jump muscle power
Sayers & Gibson, 2012	72 physically healthy and impaired older men and women Age: mean of 70.6 yrs BMI: mean of 28.8 kg.m^{-2}	High-velocity resistance exercise vs. Traditional resistance exercise vs. Stretching control group	**High-velocity resistance exercise** n= 25, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 12-14 reps at 40% of 1-RM in 2 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 2-3 sec. **Traditional resistance exercise** n= 25, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8-10 reps at 80% of 1-RM in 2 exercises. Participants were instructed to perform both concentric and eccentric phases in 2-3 sec.	NR	Leg press muscle power

Wallerstein et al., 2012	59 physically healthy older men and women Age: mean of 63.8 yrs BMI: NR	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 20, 2 sessions per week for 16 weeks performing a supervised program undertaking 2-3 sets of 4-7 reps at 30-50% of 1-RM in 6 exercises. Participants were instructed to perform the concentric phase in a high velocity. **Traditional resistance exercise** n= 20, 2 sessions per week for 16 weeks performing a supervised program undertaking 2-4 sets of 4-10 reps at 70-90% of 1-RM in 6 exercises.	NR	Leg press 1-RM
Campos et al., 2013	7 physically healthy older women Age: mean of 71.7 yrs BMI: mean of 26.7 $kg.m^{-2}$	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 4, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 6-20 RMs in 8 exercises.	NR	6-min walking test
Solberg et al., 2013	63 physically healthy older men and women Age: mean of 74.8 yrs BMI: mean of 26.3 $kg.m^{-2}$	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 33, 3 sessions per week for 13 weeks performing a supervised program undertaking 1-3 sets of 4-12 RMs in 8 exercises.	4 participants reported knee and back pain.	Fast walking speed; Timed-up and go test; Chair rise test; 6-min walking test
Yu et al., 2013	24 physically healthy older men and women Age: mean of 65.3 yrs BMI: NR	Traditional resistance exercise vs. Stretching control group	**Traditional resistance exercise** n= 12, 3 sessions per week for 5 weeks performing a supervised program in 6 exercises.	NR	Timed-up and go test
Andersen et al., 2014	17 physically healthy older men Age: mean of 68.3 yrs BMI: mean of 27.6 $kg.m^{-2}$	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 9, 2 sessions per week for 16 weeks performing a supervised program undertaking 3 sets of 8-20 RMs in 5 exercises.	NR	30-sec chair rise test
Balachandran et al., 2014	17 sarcopenic older men and women Age: mean of 71.3 yrs BMI: mean of 32.6 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 8, 2 sessions per week for 15 weeks performing a supervised program undertaking 3 sets of 10-12 reps at 50-80% of 1-RM in 6 exercises. Participants were instructed to perform	Some participants reported mild to moderate acute joint pain from the heavier weights used.	Chair rise test; Leg press 1-RM; Leg press muscle power

			the concentric phase as fast as they could and eccentric phase in 2 sec. **Traditional resistance exercise** n= 9, 2 sessions per week for 15 weeks performing a supervised program undertaking 3 sets of 10-12 reps at 70% of 1-RM in 6 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.		
Fragala et al., 2014	25 physically healthy older men and women Age: mean of 70.6 yrs BMI: mean of 28.1 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 12, 2 sessions per week for 6 weeks performing a supervised program undertaking 3 sets of 8-12 reps in 13 exercises.	NR	Timed-up and go test; Chair rise test
Lopes et al., 2014	34 physically healthy older men and women Age: mean of 65.3 yrs BMI: NR	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 10, 3 sessions per week for 12 weeks performing a supervised program undertaking 2-4 sets of 6-7 reps at 30-50% of 1-RM in 8 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 1 sec. **Traditional resistance exercise** n= 13, 3 sessions per week for 12 weeks performing a supervised program undertaking 2-4 sets of 4-10 reps at 70-90% of 1-RM in 8 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	NR	Leg press 1-RM
Pamukoff et al., 2014	20 physically healthy older men and women Age: mean of 70.8 yrs BMI: mean of 30.5 kg.m^{-2}	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 10, 3 sessions per week for 6 weeks performing a supervised program undertaking 2-3 sets of 8-10 reps at 50% of 1-RM in 7 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 2-3 sec.	Two participants reported muscle soreness during training.	Leg press 1-RM; Leg press muscle power

			Traditional resistance exercise n= 10, 3 sessions per week for 6 weeks performing a supervised program undertaking 2-3 sets of 8-10 reps at 50% of 1-RM in 7 exercises. Participants were instructed to perform both concentric and eccentric phases in 2-3 sec.		
Ramírez-Campillo et al., 2014	45 physically healthy older women Age: mean of 67.2 yrs BMI: mean of 30.7 kg.m^{-2}	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 15, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8 reps at 40-75% of 1-RM in 6 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 3 sec. **Traditional resistance exercise** n= 15, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8 reps at 40-75% of 1-RM in 6 exercises. Participants were instructed to perform both concentric and eccentric phases in 3 sec.	NR	Timed-up and go test; 30-sec chair rise test; Leg press 1-RM;
Uematsu et al., 2014	23 physically healthy older men and women Age: mean of 74.3 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 15, 2 sessions per week for 8 weeks performing a supervised program undertaking 3 sets of 10 reps at 30-40% of 1-RM in 1 exercise.	NR	Fast walking speed
Valls et al., 2014	23 physically healthy older men and women Age: mean of 72.0 yrs BMI: mean of 23.9 kg.m^{-2}	High-velocity resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 13, 2 sessions per week for 12 weeks performing a supervised program undertaking 3-4 sets of 10-15 reps at 40-70% of 1-RM in 4 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in a moderate speed.	NR	Countermovement jump muscle power;

Emerson et al., 2015	23 physically healthy older men and women Age: mean of 71.2 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 11, 2 sessions per week for 6 weeks performing a supervised program undertaking 3 sets of 8-15 reps at 70-85% of 1-RM in 11 exercises.	NR	Chair rise test
Hvid et al., 2016	37 physically impaired older men and women (mobility limitations) Age: mean of 81.9 yrs BMI: NR	High-velocity resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 16, 2 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8-10 reps at 70-80% of 1-RM in 2 exercises. Participants were instructed to perform the concentric phase as fast as they could and control the movement during the eccentric phase.	NR	Fast walking speed
Lopes et al., 2016	37 physically healthy older women Age: mean of 67.2 yrs BMI: NR	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 12, 3 sessions per week for 12 weeks performing a supervised program undertaking 3-4 sets of 6-8 reps at 40-80% of 1-RM in 7 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 2 sec. **Traditional resistance exercise** n= 14, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8 reps at 60% of 1-RM in 7 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	NR	Timed-up and go test; 30-sec chair rise test; 6-min walking test; Leg press 1-RM
Park, 2016	30 physically healthy older men Age: mean of 72.0 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 15, 3 sessions per week for 24 weeks performing a supervised program undertaking 3-5 sets of 10-15 reps at 60% of 1-RM in 14 exercises.	NR	Fast walking speed; Timed-up and go test; 30-sec chair rise test
Sayers et al., 2016	42 physically healthy or impaired older men and women	High-velocity resistance exercise vs.	**High-velocity resistance exercise** n= 28, 3 sessions per week for 12 weeks performing a supervised	NR	Fast walking speed; Timed-up and go test

	Age: mean of 71.3 yrs BMI: mean of 28.3 kg.m^{-2}	Stretching control group	program undertaking 3 sets of 14 reps at 40% of 1-RM in 2 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 2-3 sec.		
Tiggemann et al., 2016	25 physically healthy older women Age: mean of 65.6 yrs BMI: mean of 26.8 kg.m^{-2}	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 13, 2 sessions per week for 12 weeks performing a supervised program undertaking 2-3 sets of 8-15 reps at 45-65% of 1-RM in 7 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 2 sec. **Traditional resistance exercise** n= 12, 2 sessions per week for 12 weeks performing a supervised program undertaking 2-3 sets of 8-15 reps at 45-65% of 1-RM in 7 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	NR	Timed-up and go test; Chair rise test; 6-min walking test; Leg press 1-RM
Edholm et al., 2017	35 physically healthy older women Age: mean of 67.7 yrs BMI: mean of 24.7 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 17, 2 sessions per week for 24 weeks performing a supervised program undertaking 3 sets of 8-15 reps at 50-85% of 1-RM in 5 exercises.	NR	Timed-up and go test; Chair rise test
Fernández-Lezaun et al., 2017	88 physically healthy older men and women Age: mean of 69.0 yrs BMI: mean of 27.6 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 68, 1, 2 or 3 sessions per week for 24 weeks performing a supervised program undertaking 2-3 sets of 6-20 reps at 40-90% of 1-RM.	NR	Leg press 1-RM
Liao et al., 2017	46 sarcopenic older women Age: mean of 67.3 yrs BMI: mean of 27.7 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 25, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 10 reps in 6 exercises.	No adverse events related to study intervention.	Timed-up and go test; 30-sec chair rise test
Ramírez-Campillo et al., 2017	73 physically healthy older women	High-velocity resistance exercise	**High-velocity resistance exercise**	NR	Timed-up and go test;

	Age: mean of 66.6 yrs BMI: mean of 30.8 kg.m^{-2}	vs. Non-active control group	n= 58, 2 or 3 sessions per week for 12 weeks performing a supervised program undertaking 3 or 2 sets of 8 reps at 75% of 1-RM in 4 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 3 sec.		30-sec chair rise test; Leg press 1-RM
Vale et al., 2017	20 physically healthy older women Age: mean of 67.5 yrs BMI: mean of 28.6 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 10, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8-15 reps at 50-85% of 1-RM in 8 exercises.	NR	Leg press 1-RM
Filho et al., 2018	65 physically healthy older women Age: ranging from 60 to 75 yrs BMI: mean of 26.8 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 34, 3 sessions per week for 12 weeks performing a supervised program undertaking 2 sets of 15 reps in 8 exercises.	NR	Timed-up and go test; 30-sec chair rise test; 6-min walking test
Gargallo et al., 2018	93 physically healthy older women Age: mean of 70.2 yrs BMI: mean of 28.7 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 70, 2 sessions per week for 16 weeks performing a supervised program undertaking 3-4 sets of 6 reps at 85% of 1-RM or 15 reps at 70% of 1-RM in 6 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	NR	Timed-up and go test; 30-sec chair rise test; 6-min walking test
Ramírez-Campillo et al., 2018	32 physically healthy older women Age: mean of 67.3 yrs BMI: mean of 29.1 kg.m^{-2}	High-velocity resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 15, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8-12 reps at 45-75% of 1-RM in 10 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase in 3 sec.	NR	Timed-up and go test; 30-sec chair rise test
Saeterbakken et al., 2018	23 physically frail older men and women Age: mean of 84.9 yrs	High-velocity resistance exercise vs.	**High-velocity resistance exercise** n= 11, 2 sessions per week for 10 weeks performing a supervised	NR	Fast walking speed; Chair rise test;

	BMI: mean of 24.1 kg.m^{-2}	Health education control group	program undertaking 2-3 sets of 10-12 reps in 8 exercises. Participants were instructed to perform the concentric phase as fast as they could and eccentric phase slowly.		
Englund et al., 2019	70 physically impaired older men and women (mobility limitations) Age: mean of 78.9 yrs BMI: mean of 28.2 kg.m^{-2}	Traditional resistance exercise vs. Stretching control group	**Traditional resistance exercise** n= 22, 3 sessions per week for 12 weeks performing a supervised program undertaking 2-3 sets of 10-12 reps at 80% of 1-RM in 5 exercises.	NR	Leg press muscle power
Lee et al., 2019	18 physically healthy older men and women Age: mean of 66.6 yrs BMI: mean of 23.9 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 10, 2 sessions per week for 12 weeks performing a supervised program undertaking 2 sets of 10 RMs in 5 exercises.	NR	Timed-up and go test; Chair rise test; 30-sec chair rise test
Mafi et al., 2019	30 sarcopenic older men Age: mean of 68.5 yrs BMI: mean of 23.4 kg.m^{-2}	Traditional resistance exercise vs. Placebo control group	**Traditional resistance exercise** n= 14, 3 sessions per week for 8 weeks performing a supervised program undertaking 3 sets of 8-12 reps at 60-80% of 1-RM in 8 exercises.	NR	Timed-up and go test; Leg press 1-RM
Piraua et al., 2019	39 physically healthy older men and women Age: mean of 68.1 yrs BMI: mean of 28.0 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 25, 3 sessions per week for 24 weeks performing a supervised program undertaking 2-5 sets of 7-12 reps in 7 exercises.	NR	Timed-up and go test
Rashidi et al., 2019	28 physically healthy older women Age: mean of 63.1 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 20, 3 sessions per week for 8 weeks performing a supervised program undertaking the maximal number of reps at 30% or 80% of 1-RM.	NR	30-sec chair rise test
Richardson et al., 2019	50 physically healthy older men and women Age: mean of 66.2 yrs BMI: mean of 26.8 kg.m^{-2}	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 20, 1 or 2 sessions per week for 10 weeks performing a supervised program undertaking 3 sets of 14 reps at 40% of 1-RM. Participants were instructed to perform the concentric	Ten participants reported mild joint discomfort, and fifteen participants reported muscle soreness.	Timed-up and go test; 30-sec chair rise test; 6-min walking test

			phase as fast as they could and eccentric phase in 3 sec. **Traditional resistance exercise** n= 20, 1 or 2 sessions per week for 10 weeks performing a supervised program undertaking 3 sets of 7 reps at 80% of 1-RM. Participants were instructed to perform both concentric and eccentric phases in 2-3 sec.		
Urzi et al., 2019	20 physically healthy older women Age: mean of 86.4 yrs BMI: mean of 28.5 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 11, 3 sessions per week for 12 weeks performing a supervised program undertaking 8 exercises.	NR	Chair rise test
Vezzoli et al., 2019	35 sarcopenic older men and women Age: mean of 72.4 yrs BMI: mean of 27.2 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 20, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 14-16 reps at 60% of 1-RM in 5 exercises.	NR	Stair climbing muscle power
Baker et al., 2020	29 physically healthy older men and women Age: mean of 67.9 yrs BMI: mean of 32.4 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 15, 2 sessions per week for 8 weeks performing a supervised program undertaking 1-2 sets of 5-8 reps in 8 exercises. Participants were instructed to perform the concentric phase in 2 sec and eccentric phases in 4 sec.	NR	Timed-up and go test; 30-sec chair rise test
Cavalcante et al., 2020	45 physically healthy older men and women Age: mean of 71.0 yrs BMI: mean of 28.6 kg.m^{-2}	Traditional resistance exercise vs. Health education control group	**Traditional resistance exercise** n= 23, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 10-15 RM in 7 exercises. Participants were instructed to perform the concentric phase in 1 sec and eccentric phases in 2 sec.	NR	Timed-up and go test; Chair rise test; 6-min walking test; Leg press 1-RM
Choi et al., 2020	27 physically healthy older men and women Age: mean of 73.9 yrs	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 15, 3 sessions per week for 12 weeks performing a supervised program undertaking 18 exercises.	NR	Timed-up and go test; 30-sec chair rise test

	BMI: mean of 25.4 kg.m^{-2}				
Jaque et al., 2020	26 physically healthy older women Age: mean of 71.6 yrs BMI: mean of 28.5 kg.m^{-2}	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 14, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 6-25 reps in 5 exercises. Participants were instructed to perform the concentric phase as fast as they could and the eccentric phase in 1-3 sec. **Traditional resistance exercise** n= 12, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 6-25 reps in 5 exercises. Participants were instructed to perform the concentric phase in 1 sec and eccentric phases in 3 sec.	NR	Timed-up and go test; Sit-to-stand muscle power
Lopez et al., 2020	24 physically healthy older women Age: mean of 66.3 yrs BMI: mean of 25.3 kg.m^{-2}	Traditional resistance exercise vs. Walking control group	**Traditional resistance exercise** n= 12, 2 sessions per week for 8 weeks performing a supervised program undertaking 3 sets of 8-15 reps in 8 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	NR	Timed-up and go test

Ortega & Cuartas, 2020	86 physically healthy older women Age: mean of 66.7 yrs BMI: mean of 27.0 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 46, 3 sessions per week for 16 weeks performing a supervised program undertaking 4 sets up to a decrease of 10% in the set velocity of 0.6 $m.s^{-1}$. Participants were instructed to perform the concentric phase as fast as they could. **Traditional resistance exercise** n= 40, 3 sessions per week for 16 weeks performing a supervised program undertaking 4 sets at 70% of 1-RM.	NR	Fast walking speed; Timed-up and go test; 30-sec chair rise test
Van Roie et al., 2020	26 physically healthy older men Age: mean of 69.4 yrs BMI: mean of 26.6 $kg.m^{-2}$	Traditional resistance exercise vs. Walking control group	**Traditional resistance exercise** n= 12, 3 sessions per week for 12 weeks performing a supervised program undertaking 2-3 sets of 8-15 RMs in 3 exercises. Participants were instructed to perform the concentric phase in 3 sec and eccentric phases in 3 sec.	Six participants reported adverse events such as knee glutes and feet pain, and muscle soreness.	Chair rise test; 6-min walking test; Leg press 1-RM; Sit-to-stand muscle power
Coelho-Junior & Uchida, 2021	60 physically frail older men and women Age: mean of 70.1 yrs BMI: mean of 25.6 $kg.m^{-2}$	High-velocity resistance exercise vs. Traditional resistance exercise vs. Flexibility control group	**High-velocity resistance exercise** n= 22, 3 sessions per week for 16 weeks performing a supervised program undertaking 8 sets of 3-5 reps at 70-75% of 1-RM in 4 exercises. Participants were instructed to perform the concentric phase as fast as they could and the eccentric phase in 2.5 sec. **Traditional resistance exercise** n= 19, 3 sessions per week for 16 weeks performing a supervised program undertaking 4 sets of 8-10 reps at 70-75% of 1-RM in 4 exercises. Participants were instructed to perform both concentric and eccentric phases in 2.5 sec.	NR	Fast walking speed; Timed-up and go test; Chair rise test; 6-min walking test; Sit-to-stand muscle power

Lai et al., 2021	60 physically frail older men and women Age: mean of 66.5 yrs BMI: NR	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 30, 3 sessions per week for 12 weeks performing a supervised program undertaking 1 set of 10-15 reps in 5 exercises.	NR	Timed-up and go test; 30-sec chair rise test; 6-min walking test
Lee et al., 2021	293 physically healthy or frail older men and women Age: mean of 73.0 yrs BMI: mean of 23.5 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 165, 2 sessions per week for 12 weeks performing a supervised program undertaking 2 sets of 8-15 RMs in 9 exercises.	NR	Fast walking speed; Chair rise test
Lee et al., 2021 b	27 sarcopenic older women Age: mean of 70.9 yrs BMI: mean of 27.8 kg.m^{-2}	Traditional resistance exercise vs. Educational control group	**Traditional resistance exercise** n= 15, 3 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 10 reps in 5-10 exercises.	No adverse events related to study intervention.	Timed-up and go test; 30-sec chair rise test
Osuka et al., 2021	78 physically healthy or sarcopenic older women Age: mean of 71.7 yrs BMI: mean of 20.8 kg.m^{-2}	Traditional resistance exercise vs. Educational and placebo control group	**Traditional resistance exercise** n= 39, 2 sessions per week for 12 weeks performing a supervised program undertaking 1-3 sets of 8-10 reps in 10 exercises.	No adverse events related to study intervention.	Fast walking speed; Timed-up and go test; Chair rise test
Seo et al., 2021	22 sarcopenic older women Age: mean of 71.5 yrs BMI: mean of 22.7 kg.m^{-2}	Traditional resistance exercise vs. Non-active control group	**Traditional resistance exercise** n= 12, 3 sessions per week for 16 weeks performing a supervised program undertaking 3-5 sets of 6-15 reps in 17 exercises.	NR	Timed-up and go test; 30-sec chair rise test
Vieira et al., 2021	24 physically healthy older men and women Age: mean of 66.4 yrs BMI: NR	High-velocity resistance exercise vs. Traditional resistance exercise	**High-velocity resistance exercise** n= 12, 2 sessions per week for 8 weeks performing a supervised program undertaking 2 sets of 6-8 reps at 40-60% of 1-RM in 4 exercises. Participants were instructed to perform the concentric phase as fast as they could and the eccentric phase in 2 sec. **Traditional resistance exercise** n= 12, 2 sessions per week for 8 weeks performing a supervised program undertaking 2 sets of 10-12 RM in 4	NR	Timed-up and go test; 30-sec chair rise test

			exercises. Participants were instructed to perform the concentric phase in 2 sec and eccentric phases in 2 sec.		
Filho et al., 2022	54 physically healthy older men and women Age: mean of 67.0 yrs BMI: mean of 27.5 kg.m^{-2}	High-velocity resistance exercise vs. Traditional resistance exercise vs. Non-active control group	**High-velocity resistance exercise** n= 18, 2 sessions per week for 20 weeks performing a supervised program undertaking 2-3 sets of 8-12 reps in 6 exercises. Participants were instructed to perform the concentric phase as fast as they could and the eccentric phase in 2 sec. **Traditional resistance exercise** n= 17, 2 sessions per week for 20 weeks performing a supervised program undertaking 2-3 sets of 8-12 reps in 6 exercises. Participants were instructed to perform both concentric and eccentric phases in 2 sec.	NR	Timed-up and go test; 30-sec chair rise test; 6-min walking test; Leg press 1-RM
Griffen et al., 2022	18 physically healthy older women Age: mean of 67.0 yrs BMI: mean of 25.1 kg.m^{-2}	Traditional resistance exercise vs. Placebo control group	**Traditional resistance exercise** n= 9, 2 sessions per week for 12 weeks performing a supervised program undertaking 3 sets of 8-12 reps at 60-80% of 1-RM in 6 exercises.	One participant reported muscle soreness.	6-min walking test; Leg press 1-RM

1-RM, One-repetition maximum; BMI, body mass index; NR, not reported; RM, repetition maximum.

eTable 2. Individual risk of bias assessment for studies examining physical function in older adults.

Studies	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurements of the outcome	Selection of the reported results	Overall bias
Andersen et al., 2014	Some concerns	Low	Low	High	Low	High
Baker et al., 2020	Some concerns	Low	Low	Low	Low	Some concerns
Balachandran et al., 2014	Some concerns	Low	Low	Low	Some concerns	Some concerns
Bean et al., 2004	Some concerns	Low	Low	Low	Some concerns	Some concerns
Boshuizen et al., 2005	High	Low	Low	Low	Some concerns	High
Bottaro et al., 2007	High	Low	Low	High	Some concerns	High
Campos et al., 2013	High	Low	High	High	Some concerns	High
Cavalcante et al., 2020	High	Low	Low	High	Some concerns	High
Choi et al., 2020	High	Low	Low	High	Some concerns	High
Coelho-Junior & Uchida, 2021	High	Low	Low	High	Low	High
Correa et al., 2012 & Pinto et al., 2014	Some concerns	Low	Low	High	Some concerns	High
de Vreede et al., 2005	Some concerns	Low	Low	Low	Some concerns	Some concerns
Edholm et al., 2017	Some concerns	Low	Low	High	Some concerns	High
Emerson et al., 2015	Some concerns	Low	Low	High	Some concerns	High
Fahlman et al., 2007	Some concerns	Low	Low	High	Some concerns	High
Filho et al., 2018	Some concerns	Low	Low	High	Some concerns	High
Filho et al., 2022	High	Low	Low	Low	Some concerns	High
Fragala et al., 2014	Some concerns	Low	Low	High	Some concerns	High
Gargallo et al., 2018	High	Low	Low	High	Some concerns	High
Griffen et al., 2022	Low	Low	Low	Low	Low	Low
Henwood & Taaffe, 2006	Some concerns	Low	Low	High	Some concerns	High
Hvid et al., 2016	Some concerns	Low	Low	Low	Low	Some concerns
Jaque et al., 2020	Some concerns	Low	Low	Low	Some concerns	Some concerns
Judge et al., 1994	Some concerns	Low	Low	High	Some concerns	High
Kalapotharakos et al., 2010	Some concerns	Low	Low	High	Some concerns	High

Lai et al., 2021	Some concerns	Low	Low	Low	Low	Some concerns
Lee et al., 2019	High	Low	Low	High	Some concerns	High
Lee et al., 2021	Low	Low	Low	High	Low	High
Lee et al., 2021 b	Low	Low	Low	Low	Some concerns	Some concerns
Liao et al., 2017	Some concerns	Low	Low	Low	Low	Some concerns
Lopes et al., 2016	Some concerns	Low	Low	High	Some concerns	High
Lopez et al., 2020	Low	Low	Low	High	Some concerns	High
Lustosa et al., 2011	Some concerns	Low	Low	High	Some concerns	High
Mafi et al., 2019	Some concerns	Low	Low	Low	Low	Some concerns
Ortega & Cuartas, 2020	Some concerns	Low	Low	High	Some concerns	High
Osuka et al., 2021	Low	Low	Low	Low	Low	Low
Park, 2016	Some concerns	Low	Low	High	Some concerns	High
Pereira et al., 2012	Some concerns	Low	Low	High	Some concerns	High
Piraua et al., 2019	Some concerns	Low	Low	Low	Low	Some concerns
Raj et al., 2012	Some concerns	Low	Low	High	Some concerns	High
Ramírez-Campillo et al., 2014	Some concerns	Low	Low	High	Some concerns	High
Ramírez-Campillo et al., 2017	Low	Low	Low	Low	Some concerns	Some concerns
Ramírez-Campillo et al., 2018	Low	Low	Low	Low	Some concerns	Some concerns
Rashidi et al., 2019	High	Low	Low	High	Some concerns	High
Richardson et al., 2019	High	Low	Low	High	Some concerns	High
Saeterbakken et al., 2018	High	Low	Low	High	Some concerns	High
Sayers et al., 2016	Low	Low	Low	Low	Some concerns	Some concerns
Seo et al., 2021	High	Low	Low	High	Some concerns	High
Sipilä et al., 1996	Some concerns	Low	Low	High	Some concerns	High
Solberg et al., 2013	High	Low	Low	Low	Some concerns	High
Sousa & Sampaio, 2005	Some concerns	Low	Low	High	Some concerns	High
Tiggemann et al., 2016	Some concerns	Low	Low	Low	Some concerns	Some concerns
Uematsu et al., 2014	Some concerns	Low	Low	High	Some concerns	High
Urzi et al., 2019	Some concerns	Low	Low	High	Low	High
Van Roie et al., 2020	High	Low	Low	High	Some concerns	High
Vieira et al., 2021	Some concerns	Low	Low	High	Low	High

Yu et al., 2013	Some concerns	Low	Low	High	Some concerns	High

eTable 3. Individual risk of bias assessment for studies examining muscle power in older adults.

Studies	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurements of the outcome	Selection of the reported results	Overall bias
Muscle power						
Balachandran et al., 2014	Some concerns	Low	Low	Low	Some concerns	Some concerns
Bean et al., 2004	Some concerns	Low	Low	Low	Some concerns	Some concerns
Bean et al., 2009	Some concerns	Low	Low	Low	Some concerns	Some concerns
Bottaro et al., 2007	High	Low	Low	High	Some concerns	High
Caserotti et al., 2008	Some concerns	Low	Low	High	Some concerns	High
Coelho-Junior & Uchida, 2021	Some concerns	Low	Low	High	Low	High
de Vos et al., 2005	Some concerns	Low	Low	High	Some concerns	High
de Vreede et al., 2005	Some concerns	Low	Low	Low	Some concerns	Some concerns
Englund et al., 2019	Some concerns	Low	Low	Low	Low	Some concerns
Fielding et al., 2002	High	Low	Low	High	Some concerns	High
Henwood & Taaffe, 2006	Some concerns	Low	Low	High	Some concerns	High
Jaque et al., 2020	Some concerns	Low	Low	Low	Some concerns	Some concerns
Marsh et al., 2009	High	Low	Low	High	Some concerns	High
Pamukoff et al., 2014	Some concerns	Low	Low	High	Some concerns	High
Raj et al., 2012	Some concerns	Low	Low	High	Some concerns	High
Reid et al., 2008	Some concerns	Low	Low	High	Some concerns	High
Sayers & Gibson, 2012	High	Low	Low	High	Some concerns	High
Valls et al., 2014	Some concerns	Low	Low	High	Some concerns	High
Van Roie et al., 2020	Some concerns	Low	Low	High	Some concerns	High
Vezzoli et al., 2019	Some concerns	Low	Low	High	Some concerns	High

eTable 4. Individual risk of bias assessment for studies examining muscle strength in older adults.

Studies	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurements of the outcome	Selection of the reported results	Overall bias
Balachandran et al., 2014	Some concerns	Low	Low	Low	Some concerns	Some concerns
Bottaro et al., 2007	High	Low	Low	High	Some concerns	High
Cavalcante et al., 2020	High	Low	Low	High	Some concerns	High
Charette et al., 1991	High	Low	Low	High	Some concerns	High
Fatouros et al., 2002	Some concerns	Low	Low	High	Some concerns	High
Fernández-Lezaun et al., 2017	High	Low	Low	High	Some concerns	High
Filho et al., 2022	High	Low	Low	Low	Some concerns	High
Griffen et al., 2022	Low	Low	Low	Low	Low	Low
Haykowsky et al., 2000	High	Low	Low	High	Some concerns	High
Henwood & Taaffe, 2006	Some concerns	Low	Low	High	Some concerns	High
Lopes et al., 2014	High	Low	Low	High	Some concerns	High
Lopes et al., 2016	Some concerns	Low	Low	High	Some concerns	High
Mafi et al., 2019	Some concerns	Low	Low	Low	Low	Some concerns
McCartney et al., 1995	High	Low	Low	High	Some concerns	High
Miszko et al., 2003	High	Low	Low	High	Some concerns	High
Pamukoff et al., 2014	High	Low	Low	High	Some concerns	High
Ramírez-Campillo et al., 2014	Some concerns	Low	Low	High	Some concerns	High
Ramírez-Campillo et al., 2017	Low	Low	Low	Low	Some concerns	Some concerns
Rhodes et al., 2000	Some concerns	Low	Low	High	Some concerns	High
Taaffe et al., 1999	High	Low	Low	High	Some concerns	High
Tiggemann et al., 2016	Some concerns	Low	Low	Low	Some concerns	Some concerns
Vale et al., 2017	Some concerns	Low	Low	High	Some concerns	High
Van Roie et al., 2020	High	Low	Low	High	Some concerns	High
Wallerstein et al., 2012	High	Low	Low	High	Some concerns	High

eTable 5. Pairwise meta-analysis on fast walking speed.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1		Group 2		Between-group SMD (95% CI)
						n	MD ± SD	n	MD ± SD	
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	-0.05 ± 0.80	9	0.03 ± 0.47	-0.11 (95% CI: -1.00 to 0.77)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-0.10 ± 0.26	10	0.20 ± 0.36	-0.92 (95% CI: -1.83 to -0.01)
Hvid et al., 2016 *	PI	37.8%	81.9	NR	Some concerns	16	0.09 ± 0.26	21	-0.03 ± 0.18	0.53 (95% CI: -0.13 to 1.20)
Saeterbakken et al., 2018 *	PI	26.1%	84.9	24.1	High risk	11	0.09 ± 0.10	12	-0.06 ± 0.11	1.38 (95% CI: 0.45 to 2.31)
Sayers et al., 2016	PH and PI	38.1%	71.3	28.3	Some concerns	28	-0.02 ± 0.90	14	-0.06 ± 0.37	0.05 (95% CI: -0.59 to 0.69)
Overall effect					***High-velocity resistance exercise vs. control group; I^2= 71%, τ^2= 0.39***					***0.19 (95% CI: -0.46 to 0.84)***
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	8	0.02 ± 0.42	9	0.03 ± 0.47	-0.02 (95% CI: -0.97 to 0.93)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	0.00 ± 0.30	10	0.20 ± 0.36	-0.58 (95% CI: -1.46 to 0.30)
Judge et al., 1994 *	PH	56.4%	80.5	NR	High risk	28	0.02 ± 3.70	27	0.03 ± 1.48	0.00 (95% CI: -0.53 to 0.53)
Lee et al., 2021	PH and PI	21.5%	73.0	23.5	Some concerns	132	0.11 ± 0.26	118	0.02 ± 0.30	0.32 (95% CI: 0.07 to 0.57)
Osuka et al., 2021	PH and PI	0.0%	71.7	20.8	Low risk	39	0.22 ± 0.22	39	0.07 ± 0.19	0.71 (95% CI: 0.26 to 1.17)

Park, 2016 *	PH	100.0%	72.0	NR	High risk	15	0.09 ± 0.26	15	-0.03 ± 0.30	0.41 (95% CI: -0.31 to 1.14)
Sipila et al., 1996 *	PH	0.0%	NR	NR	High risk	12	0.19 ± 0.27	11	0.09 ± 0.21	0.40 (95% CI: -0.43 to 1.23)
Solberg et al., 2013	PH	30.2%	74.8	26.3	High risk	30	0.07 ± 0.24	33	0.02 ± 0.25	0.20 (95% CI: -0.29 to 0.70)
Uematsu et al., 2014 *	PH	52.2%	74.3	NR	High risk	15	0.10 ± 0.23	8	0.13 ± 0.15	-0.14 (95% CI: -1.00 to 0.72)
Overall effect	***Traditional resistance exercise vs. control group, I^2= 20%, τ^2= 0.02***									***0.25 (95% CI: 0.04 to 0.46)***
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	-0.05 ± 0.80	8	0.02 ± 0.42	-0.10 (95% CI: -1.01 to 0.81)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-0.10 ± 0.26	11	0.00 ± 0.30	-0.34 (95% CI: -1.18 to 0.50)
Ortega & Cuartas, 2020 *	PH	0.0%	66.7	27.0	High risk	46	0.40 ± 0.17	40	0.19 ± 0.20	1.12 (95% CI: 0.66 to 1.58)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 83%, τ^2= 0.67***									***0.28 (95% CI: 0.74 to 1.31)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 6. Pairwise meta-analysis on timed up and go.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1		Group 2		Between-group SMD (95% CI)
						n	MD ± SD	n	MD ± SD	
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	3.10 ± 24.51	9	2.30 ± 36.86	0.03 (95% CI: -0.86 to 0.91)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-0.80 ± 2.61	10	0.10 ± 1.40	-0.41 (95% CI: -1.27 to 0.46)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	-1.75 ± 1.15	19	-0.30 ± 2.85	-0.65 (95% CI: -1.31 to 0.02)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	-0.63 ± 0.89	11	0.03 ± 0.68	-0.80 (95% CI: -1.65 to 0.06)
Pereira et al., 2012 *	PH	0.0%	62.5	27.6	High risk	28	-0.71 ± 0.91	28	0.21 ± 0.70	-1.12 (95% CI: -1.69 to -0.55)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	-1.40 ± 1.28	15	0.10 ± 1.35	-1.11 (95% CI: -1.89 to -0.33)
Ramírez-Campillo et al., 2017 *	PH	0.0%	66.6	30.8	Some concerns	58	-0.95 ± 1.41	15	0.10 ± 1.35	-0.74 (95% CI: -1.33 to -0.16)
Ramírez-Campillo et al., 2018 *	PH	0.0%	67.3	29.1	Some concerns	15	-0.70 ± 1.30	17	0.10 ± 1.50	-0.55 (95% CI: -1.26 to 0.16)
Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	-0.40 ± 1.19	10	0.10 ± 0.66	-0.46 (95% CI: -1.23 to 0.31)
Sayers et al., 2016 *	PH and PI	38.1%	71.3	28.3	High risk	28	-0.50 ± 6.51	14	0.00 ± 3.07	-0.09 (95% CI: -0.73 to 0.55)
Overall effect	***High-velocity resistance exercise vs. control group; I^2= 10%, τ^2= 0.01***									***-0.62 (95% CI: -0.86 to -0.39)***

Baker et al., 2020 *	PH	34.0%	67.9	32.4	Some concerns	13	-0.86 ± 0.43	16	-0.77 ± 0.38	-0.22 (95% CI: -0.95 to 0.52)
Boshuizen et al., 2005 *	PI	6.1%	78.6	NR	High risk	16	-1.30 ± 7.50	17	0.10 ± 9.62	-0.16 (95% CI: -0.84 to 0.53)
Cavalcante et al., 2020 *	PH	26.7%	71.0	28.6	High risk	23	-1.90 ± 1.83	22	-0.20 ± 1.14	-1.09 (95% CI: -1.72 to -0.46)
Choi et al., 2020 *	PH	NR	73.9	25.4	High risk	15	-5.10 ± 12.70	12	-0.10 ± 2.27	-0.50 (95% CI: -1.28 to 0.27)
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	8	-55.60 ± 161.79	9	2.30 ± 36.86	-0.48 (95% CI: -1.45 to 0.49)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-0.50 ± 0.80	10	0.10 ± 1.40	-0.51 (95% CI: -1.39 to 0.36)
De Vreedes et al., 2005	PH	0.0%	73.9	NR	Some concerns	34	-0.10 ± 0.70	31	0.10 ± 0.70	-0.28 (95% CI: -0.77 to 0.21)
Edholm et al., 2017 *	PH	0.0%	67.7	24.7	High risk	17	-1.19 ± 1.44	18	-1.05 ± 2.24	-0.07 (95% CI: -0.74 to 0.59)
Filho et al., 2018 *	PH	0.0%	NR	26.8	High risk	34	-1.62 ± 1.43	31	0.00 ± 1.39	-1.14 (95% CI: -1.66 to -0.61)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	17	-2.12 ± 1.14	19	-0.30 ± 2.85	-0.80 (95% CI: -1.49 to -0.12)
Fragala et al., 2014 *	PH	56.5%	70.6	28.1	High risk	12	-1.15 ± 2.23	13	-0.62 ± 0.90	-0.31 (95% CI: -1.10 to 0.48)
Gargallo et al., 2018 *	PH	0.0%	70.2	28.7	High risk	70	-0.87 ± 0.88	23	-0.25 ± 1.69	-0.55 (95% CI: -1.03 to -0.07)
Kalapotharakos et al., 2010 *	PH	100.0%	83.0	NR	High risk	7	-2.03 ± 2.51	7	0.24 ± 1.65	-1.00 (95% CI: -2.14 to 0.13)

Lai et al., 2021 *	PI	53.3%	66.5	NR	High risk	30	-1.09 ± 0.82	30	-0.22 ± 0.60	-1.20 (95% CI: -1.75 to -0.65)
Lee et al., 2019 *	PH	55.6%	66.6	23.9	High risk	10	-0.50 ± 0.66	8	0.40 ± 0.60	-1.36 (95% CI: -2.41 to -0.30)
Lee et al., 2021 b *	PI	0.0%	70.9	27.8	Some concerns	15	-1.95 ± 8.07	12	0.27 ± 10.59	-0.23 (95% CI: -0.99 to 0.53)
Liao et al., 2017	PI	0.0%	67.3	27.7	Some concerns	25	-1.39 ± 1.20	21	0.03 ± 1.19	-1.17 (95% CI: -1.80 to -0.54)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	14	0.05 ± 0.79	11	0.03 ± 0.68	0.03 (95% CI: -0.76 to 0.82)
Lopez et al., 2020	PH	0.0%	66.3	25.3	High risk	12	-0.14 ± 0.89	12	0.16 ± 0.57	-0.39 (95% CI: -1.20 to 0.42)
Lustosa et al., 2011 *	PI	0.0%	72.0	29.1	High risk	32	-0.68 ± 2.13	16	-0.72 ± 2.14	0.02 (95% CI: -0.58 to 0.62)
Mafi et al., 2019	PI	100.0%	68.5	23.4	Some concerns	14	-0.22 ± 0.21	16	0.14 ± 0.23	-1.59 (95% CI: -2.42 to -0.75)
Osuka et al., 2021	PH and PI	0.0%	71.7	20.8	Low risk	39	-0.80 ± 0.64	39	-0.40 ± 0.64	-0.62 (95% CI: -1.08 to -0.17)
Park, 2016 *	PH	100.0%	72.0	NR	High risk	15	-0.22 ± 0.98	15	0.02 ± 1.00	-0.24 (95% CI: -0.95 to 0.48)
Pinto et al., 2014 *	PH	0.0%	67.0	NR	High risk	19	-1.07 ± 0.44	17	0.10 ± 0.54	-2.33 (95% CI: -3.20 to -1.47)
Piraua et al., 2019 *	PH	7.7%	68.1	28.0	Some concerns	25	-0.46 ± 0.65	14	1.43 ± 0.58	-2.94 (95% CI: -3.89 to -1.99)
Raj et al., 2012 *	PH	0.0%	67.5	NR	High risk	12	-0.21 ± 0.74	10	-0.08 ± 0.51	-0.19 (95% CI: -1.03 to 0.65)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	-0.80 ± 1.35	15	0.10 ± 1.35	-0.65 (95% CI: -1.38 to 0.09)

Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	-0.30 ± 0.67	10	0.10 ± 0.66	-0.59 (95% CI: -1.36 to 0.19)
Seo et al., 2021 *	PI	0.0%	71.5	22.7	High risk	12	-0.60 ± 0.45	10	0.40 ± 0.72	-1.64 (95% CI: -2.64 to -0.65)
Solberg et al., 2013	PH	30.2%	74.8	26.3	High risk	30	-0.20 ± 0.70	33	-0.10 ± 0.59	-0.15 (95% CI: -0.65 to 0.34)
Sousa & Sampaio, 2005 *	PH	100.0%	74.0	NR	High risk	10	-0.90 ± 1.26	10	0.50 ± 3.65	-0.49 (95% CI: -1.38 to 0.40)
Yu et al., 2013 *	PH	58.3%	65.3	NR	High risk	12	-0.10 ± 0.98	12	-0.20 ± 1.15	0.09 (95% CI: -0.71 to 0.89)
Overall effect					***Traditional resistance exercise vs. control group, I^2= 39%, τ^2= 0.08***					***-0.57 (95% CI: -0.75 to -0.39)***
Bottaro et al., 2007 *	PH	100.0%	66.5	21.2	High risk	11	-0.89 ± 0.87	9	-0.04 ± 0.68	-1.03 (95% CI: -1.98 to -0.08)
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	3.10 ± 24.51	8	-55.60 ± 161.79	0.53 (95% CI: -0.40 to 1.46)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-0.80 ± 2.61	11	-0.50 ± 0.80	-0.15 (95% CI: -0.99 to 0.69)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	-1.75 ± 1.15	17	-2.12 ± 1.14	0.32 (95% CI: -0.35 to 0.98)
Jaque et al., 2020 *	PH	0.0%	71.6	28.5	Some concerns	14	-1.07 ± 0.84	12	0.17 ± 0.97	-1.33 (95% CI: -2.20 to -0.47)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	-0.63 ± 0.89	14	0.05 ± 0.79	-0.78 (95% CI: -1.59 to 0.02)
Ortega & Cuartas, 2020 *	PH	0.0%	66.7	27.0	High risk	46	-1.28 ± 0.99	40	-0.97 ± 0.87	-0.33 (95% CI: -0.75 to 0.10)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	-1.40 ± 1.28	15	-0.80 ± 1.35	-0.44 (95% CI: -1.17 to 0.28)

Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	-0.40 ± 1.19	20	-0.30 ± 0.67	-0.10 (95% CI: -0.72 to 0.52)
Tiggemann et al., 2016 *	PH	0.0%	65.6	26.8	Some concerns	13	-0.40 ± 0.40	12	-0.70 ± 0.66	0.54 (95% CI: -0.26 to 1.34)
Vieira et al., 2021 *	PH	29.2%	66.4	NR	High risk	12	-0.44 ± 1.04	12	-0.51 ± 0.43	0.08 (95% CI: -0.72 to 0.89)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 52%, τ^2= 0.15***									***-0.23 (95% CI: -0.55 to 0.09)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 7. Pairwise meta-analysis on five times sit-to-stand test.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1		Group 2		Between-group SMD (95% CI)
						n	MD ± SD	n	MD ± SD	
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	-7.30 ± 12.00	9	8.50 ± 16.76	-1.06 (95% CI: -2.01 to -0.10)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-2.70 ± 2.17	10	1.00 ± 0.95	-2.09 (95% CI: -3.19 to -0.98)
Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	23	-1.40 ± 1.79	22	0.00 ± 1.71	-0.79 (95% CI: -1.40 to -0.18)
Saeterbakken et al., 2018 *	PI	26.1%	84.9	24.1	High risk	11	-2.10 ± 4.37	12	-1.50 ± 2.25	-0.17 (95% CI: -0.99 to 0.65)
Overall effect	***High-velocity resistance exercise vs. control group; I^2= 61%, τ^2= 0.29***									***-0.95 (95% CI: -1.62 to -0.27)***
Cavalcante et al., 2020 *	PH	26.7%	71.0	28.6	High risk	23	-4.40 ± 4.59	22	-1.50 ± 3.27	-0.71 (95% CI: -1.32 to -0.11)
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	8	-9.60 ± 11.65	9	8.50 ± 16.76	-1.18 (95% CI: -2.23 to -0.12)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-1.80 ± 0.98	10	1.00 ± 0.95	-2.77 (95% CI: -4.03 to 1.51)
Edholm et al., 2017 *	PH	0.0%	67.7	24.7	High risk	17	-3.32 ± 1.49	18	-3.32 ± 0.94	0.00 (95% CI: -0.66 to 0.66)
Emerson et al., 2015 *	PH	NR	71.2	NR	High risk	11	-3.10 ± 2.63	12	-0.80 ± 1.93	-0.97 (95% CI: -1.84 to -0.09)
Fragala et al., 2014 *	PH	56.5%	70.6	28.1	High risk	12	-2.62 ± 2.47	13	-0.78 ± 1.95	-0.80 (95% CI: -1.62 to 0.02)

Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	22	-0.70 ± 2.01	22	0.00 ± 1.71	-0.37 (95% CI: -0.97 to 0.23)
Kalapotharakos et al., 2010 *	PH	100.0%	83.0	NR	High risk	7	-2.45 ± 1.91	7	0.07 ± 2.03	-1.20 (95% CI: -2.37 to -0.03)
Lee et al., 2019 *	PH	55.6%	66.6	23.9	High risk	10	-0.60 ± 1.47	8	0.20 ± 1.00	-0.59 (95% CI: -1.55 to 0.36)
Lee et al., 2021	PH and PI	21.5%	73.0	23.5	High risk	132	-1.20 ± 3.90	118	-0.40 ± 4.10	-0.20 (95% CI: -0.45 to 0.05)
Osuka et al., 2021	PH and PI	0.0%	71.7	20.8	Low risk	39	-2.00 ± 1.12	39	-0.70 ± 0.80	-1.33 (95% CI: -1.82 to -0.84)
Solberg et al., 2013	PH	30.2%	74.8	26.3	High risk	30	-1.00 ± 1.26	33	-0.60 ± 1.17	-0.33 (95% CI: -0.82 to 0.17)
Urzi et al., 2019 *	PH	0.0%	86.4	28.5	High risk	11	-7.30 ± 4.91	9	-0.10 ± 9.95	-0.91 (95% CI: -1.85 to 0.03)
Van Roie et al., 2020 *	PH	100.0%	69.4	26.6	High risk	12	0.46 ± 1.45	14	0.13 ± 1.46	0.22 (95% CI: -0.55 to 0.99)
Overall effect					***Traditional resistance exercise vs. control group, I^2= 68%, τ^2= 0.22***					***-0.68 (95% CI: -1.00 to -0.37)***
Balachandran et al., 2014 *	PI	5.9%	71.3	32.6	Some concerns	8	-2.20 ± 6.31	9	-2.70 ± 5.05	0.08 (95% CI: -0.87 to 1.04)
Bean et al., 2004	PI	0.0%	78.0	26.3	Some concerns	11	-8.20 ± 2.50	13	-5.70 ± 1.60	-1.17 (95% CI: -2.05 to -0.29)
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	-7.30 ± 12.00	8	-9.60 ± 11.65	0.19 (95% CI: -0.73 to 1.10)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	-2.70 ± 2.17	11	-1.80 ± 0.98	0.51 (95% CI: -1.37 to 0.34)

Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	23	-1.40 ± 1.79	22	-0.70 ± 2.01	-0.36 (95% CI: -0.95 to 0.23)
Tiggemann et al., 2016 *	PH	0.0%	65.6	26.8	Some concerns	13	-1.80 ± 1.31	12	-2.90 ± 1.54	0.75 (95% CI: -0.07 to 1.56)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 61%, τ^2= 0.28***									***-0.02 (95% CI: -0.55 to 0.53)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 8. Pairwise meta-analysis on 30-sec sit-to-stand.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1		Group 2		Between-group SMD (95% CI)
						n	MD ± SD	n	MD ± SD	
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	-1.41 ± 2.37	19	-0.38 ± 1.66	-0.50 (95% CI: -1.15 to 0.16)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	2.66 ± 1.98	11	0.17 ± 3.03	0.95 (95% CI: 0.08 to 1.82)
Pereira et al., 2012 *	PH	0.0%	62.5	27.6	High risk	28	4.38 ± 5.54	28	-1.97 ± 6.28	1.06 (95% CI: 0.50 to 1.62)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	2.60 ± 2.93	15	-0.10 ± 2.15	1.02 (95% CI: 0.25 to 1.79)
Ramírez-Campillo et al., 2017 *	PH	0.0%	66.6	30.8	Some concerns	58	2.05 ± 3.22	15	-0.10 ± 2.15	0.70 (95% CI: 0.12 to 1.28)
Ramírez-Campillo et al., 2018 *	PH	0.0%	67.3	29.1	Some concerns	15	1.80 ± 2.98	17	0.40 ± 2.31	0.52 (95% CI: -0.19 to 1.22)
Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	1.00 ± 4.19	10	0.00 ± 2.00	0.27 (95% CI: -0.49 to 1.03)
Overall effect		***High-velocity resistance exercise vs. control group; I^2= 62%, τ^2= 0.20***								***0.57 (95% CI: 0.15 to 0.99)***
Andersen et al., 2014 *	PH	100.0%	68.3	27.6	High risk	9	5.00 ± 1.73	8	0.00 ± 0.00	3.75 (95% CI: 2.03 to 5.48)
Baker et al., 2020 *	PH	34.0%	67.9	32.4	Some concerns	13	2.40 ± 1.18	16	0.39 ± 0.34	2.37 (95% CI: 1.38 to 3.35)
Choi et al., 2020 *	PH	NR	73.9	25.4	High risk	15	3.60 ± 5.03	12	1.50 ± 3.17	0.47 (95% CI: -0.30 to 1.24)
Correa et al., 2012 *	PH	0.0%	67.0	NR	High risk	42	3.10 ± 1.41	17	0.00 ± 2.00	1.91 (95% CI: 1.25 to 2.58)
Fahlman et al., 2007 *	PI	NR	75.5	NR	High risk	39	3.00 ± 4.88	33	1.40 ± 4.02	0.35 (95% CI: -0.12 to 0.82)
Filho et al., 2018 *	PH	0.0%	NR	26.8	High risk	34	3.50 ± 2.29	31	0.50 ± 1.87	1.41 (95% CI: 0.87 to 1.96)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	17	3.34 ± 1.90	19	-0.38 ± 1.66	2.05 (95% CI: 1.22 to 2.87)

Gargallo et al., 2018 *	PH	0.0%	70.2	28.7	High risk	70	6.78 ± 3.17	23	1.46 ± 6.38	1.26 (95% CI: 0.76 to 1.77)
Lai et al., 2021 *	PI	53.3%	66.5	NR	Some concerns	30	4.34 ± 3.67	30	0.27 ± 4.16	1.02 (95% CI: 0.48 to 1.56)
Lee et al., 2019 *	PH	55.6%	66.6	23.9	High risk	10	2.20 ± 4.55	8	-1.00 ± 2.01	0.83 (95% CI: -0.15 to 1.81)
Lee et al., 2021 b *	PI	0.0%	70.9	27.8	Some concerns	15	1.66 ± 13.54	12	-1.25 ± 17.68	0.18 (95% CI: -0.58 to 0.94)
Liao et al., 2017	PI	0.0%	67.3	27.7	Some concerns	25	3.53 ± 3.45	21	-1.02 ± 3.48	1.29 (95% CI: 0.65 to 1.93)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	14	0.63 ± 2.17	11	0.17 ± 3.03	0.17 (95% CI: -0.62 to 0.96)
Park, 2016 *	PH	100.0%	72.0	NR	High risk	15	1.90 ± 3.26	15	-0.20 ± 3.34	0.62 (95% CI: -0.12 to 1.35)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	1.70 ± 3.40	15	-0.10 ± 2.15	0.61 (95% CI: -0.12 to 1.35)
Rashidi et al., 2019 *	PH	0.0%	63.1	NR	High risk	20	5.25 ± 3.00	8	1.75 ± 2.52	1.18 (95% CI: 0.29 to 2.06)
Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	2.00 ± 2.58	10	0.00 ± 2.00	0.81 (95% CI: 0.02 to 1.60)
Seo et al., 2021 *	PI	0.0%	71.5	22.7	High risk	12	4.80 ± 4.86	10	0.00 ± 3.11	1.11 (95% CI: 0.19 to 2.02)
Overall effect		***Traditional resistance exercise vs. control group, I^2= 69%, τ^2= 0.29***								***1.09 (95% CI: 0.78 to 1.40)***
Bottaro et al., 2007 *	PH	100.0%	66.5	21.2	High risk	11	7.63 ± 5.53	9	1.33 ± 3.46	1.28 (95% CI: 0.29 to 2.26)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	-1.41 ± 2.37	17	3.34 ± 1.90	-2.15 (95% CI: -3.00 to -1.30)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	2.66 ± 1.98	14	0.63 ± 2.17	0.94 (95% CI: 0.12 to 1.76)
Ortega & Cuartas, 2020 *	PH	0.0%	66.7	27.0	High risk	46	13.05 ± 6.70	40	8.08 ± 5.14	0.82 (95% CI: 0.38 to 1.26)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	2.60 ± 2.93	15	1.70 ± 3.40	0.28 (95% CI: -0.44 to 1.00)

Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	1.00 ± 4.19	20	2.00 ± 2.58	-0.28 (95% CI: -0.91 to 0.34)
Vieira et al., 2021 *	PH	29.2%	66.4	NR	High risk	12	1.85 ± 1.66	12	2.51 ± 1.81	-0.37 (95% CI: -1.18 to 0.44)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 88%, τ^2= 0.92***									***0.08 (95% CI: -0.69 to 0.84)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 9. Pairwise meta-analysis on 6-min walking test.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1		Group 2		Between-group SMD (95% CI)
						n	MD ± SD	n	MD ± SD	
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	18.00 ± 155.15	10	0.00 ± 179.00	0.10 (95% CI: -0.75 to 0.96)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	142.67 ± 64.87	19	5.09 ± 80.08	1.84 (95% CI: 1.06 to 2.62)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	-6.13 ± 63.26	11	3.06 ± 55.10	-0.15 (95% CI: -0.97 to 0.67)
Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	34.00 ± 92.81	10	15.00 ± 81.91	0.21 (95% CI: -0.55 to 0.97)
Overall effect		***High-velocity resistance exercise vs. control group; I^2= 80%, τ^2= 0.68***								***0.51 (95% CI: -0.39 to 1.41)***
Campos et al., 2013 *	PH	0.0%	71.7	26.7	High risk	4	6.20 ± 50.29	3	-7.40 ± 33.94	0.26 (95% CI: -1.25 to 1.77)
Cavalcante et al., 2020 *	PH	26.7%	71.0	28.6	High risk	23	101.00 ± 63.66	22	13.00 ± 53.51	1.47 (95% CI: 0.80 to 2.13)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	31.00 ± 136.01	10	0.00 ± 179.00	0.19 (95% CI: -0.67 to 1.05)
Fahlman et al., 2007 *	PI	NR	75.5	NR	High risk	39	35.67 ± 124.11	33	43.89 ± 126.09	-0.07 (95% CI: -0.53 to 0.40)
Filho et al., 2018 *	PH	0.0%	NR	26.8	High risk	34	60.95 ± 48.50	31	-5.00 ± 40.88	1.45 (95% CI: 0.90 to 2.00)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	17	132.49 ± 126.04	19	5.09 ± 80.08	1.19 (95% CI: 0.48 to 1.91)

Gargallo et al., 2018 *	PH	0.0%	70.2	28.7	High risk	70	23.99 ± 55.11	23	12.81 ± 100.17	0.16 (95% CI: -0.31 to 0.63)
Griffen et al., 2022	PH	0.0%	67.0	25.1	Low risk	9	20.71 ± 11.27	9	-3.42 ± 12.58	1.92 (95% CI: 0.76 to 3.09)
Kalapotharakos et al., 2010 *	PH	100.0%	83.0	NR	High risk	7	90.44 ± 58.36	7	-3.87 ± 38.07	1.79 (95% CI: 0.48 to 3.10)
Lai et al., 2021 *	PI	53.3%	66.5	NR	Some concerns	30	69.14 ± 67.07	30	-10.88 ± 65.52	1.19 (95% CI: 0.64 to 1.74)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	14	0.00 ± 59.06	11	3.06 ± 55.10	-0.05 (95% CI: -0.84 to 0.74)
Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	42.00 ± 86.78	10	15.00 ± 81.91	0.31 (95% CI: -0.46 to 1.07)
Solberg et al., 2013	PH	30.2%	74.8	26.3	High risk	30	11.00 ± 51.70	33	-2.00 ± 54.22	0.24 (95% CI: -0.25 to 0.74)
Van Roie et al., 2020 *	PH	100.0%	69.4	26.6	High risk	12	39.30 ± 79.55	14	27.00 ± 79.38	0.15 (95% CI: -0.62 to 0.92)
Overall effect					***Traditional resistance exercise vs. control group, I^2= 74%, τ^2= 0.33***					***0.68 (95% CI: 0.32 to 1.05)***
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	18.00 ± 155.15	11	31.00 ± 136.01	-0.09 (95% CI: -0.92 to 0.75)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	142.67 ± 64.87	17	132.49 ± 126.04	0.10 (95% CI: -0.56 to 0.76)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	-6.13 ± 63.26	14	0.00 ± 59.06	-0.10 (95% CI: -0.87 to 0.67)
Richardson et al., 2019 *	PH	50.0%	66.2	26.8	High risk	20	34.00 ± 92.81	20	42.00 ± 86.78	-0.09 (95% CI: -0.71 to 0.53)

Tiggemann et al., 2016 *	PH	0.0%	65.6	26.8	Some concerns	13	48.10 ± 45.01	12	38.60 ± 36.98	0.22 (95% CI: -0.57 to 1.01)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 0%, τ^2= 0.00***									***0.01 (95% CI: -0.32 to 0.33)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 10. Side-split analyses from network meta-analysis on physical function outcomes.

Comparisons	Direct comparison SMD (95% CI)	Indirect comparison SMD (95% CI)	Difference in estimates SMD (95% CI)	P-value
Fast walking speed				
HVRE vs. Control groups	0.19 (-0.31 to 0.69)	1.21 (0.33 to 2.10)	-1.02 (-2.04 to -0.00)	**0.049**
TRE vs. Control groups	0.20 (-0.14 to 0.53)	-0.18 (-1.25 to 0.88)	0.38 (-0.74 to 1.49)	0.508
HVRE vs. TRE	0.39 (-0.22 to 1.00)	0.13 (-0.57 to 0.83)	0.26 (-0.68 to 1.19)	0.589
Timed-up and go				
HVRE vs. Control groups	-0.61 (-0.96 to -0.25)	-1.07 (-1.58 to -0.56)	0.47 (-0.16 to 1.09)	0.144
TRE vs. Control groups	-0.67 (-0.87 to -0.47)	-0.22 (-0.96 to 0.52)	-0.45 (-1.22 to 0.31)	0.246
HVRE vs. TRE	-0.23 (-0.58 to 0.12)	-0.14 (-0.39 to 0.67)	-0.38 (-1.01 to 0.26)	0.245
Five times sit-to-stand				
HVRE vs. Control groups	-0.93 (-1.54 to -0.32)	-0.47 (-1.19 to 0.24)	-0.46 (-1.40 to 0.49)	0.343
TRE vs. Control groups	-0.68 (-0.99 to -0.37)	-0.24 (-1.56 to 1.09)	-0.44 (-1.80 to 0.92)	0.526
HVRE vs. TRE	-0.01 (-0.51 to 0.48)	-0.34 (-1.30 to 0.62)	0.33 (-0.75 to 1.41)	0.553
30-sec sit-to-stand				
HVRE vs. Control groups	0.57 (0.03 to 1.11)	1.67 (0.82 to 2.52)	-1.10 (-2.10 to -0.09)	**0.032**
TRE vs. Control groups	1.10 (0.75 to 1.44)	-0.01 (-1.16 to 1.14)	1.11 (-0.09 to 2.31)	0.070
HVRE vs. TRE	0.08 (-0.46 to 0.63)	-0.60 (-1.43 to 0.23)	0.68 (-0.31 to 1.67)	0.179
6-min walking test				
HVRE vs. Control groups	0.51 (-0.15 to 1.17)	0.91 (-0.12 to 1.93)	-0.39 (-1.61 to 0.82)	0.526
TRE vs. Control groups	0.68 (0.33 to 1.03)	0.86 (-2.20 to 3.91)	-0.17 (-3.25 to 2.90)	0.912
HVRE vs. TRE	0.01 (-0.56 to 0.58)	-0.47 (-1.91 to 0.97)	0.48 (-1.06 to 2.03)	0.540

95% CI, 95% confidence interval; HVRE, high-velocity resistance exercise; TRE, traditional resistance exercise; SMD, standardised mean difference.

eTable 11. Sensitivity analyses by omitting outliers from network meta-analysis on physical function outcomes.

Comparisons	k	ES	95% CI	P-value	I^2	Best intervention probability
Fast walking speed						
HVRE vs. Control groups	5	0.24	-0.16 to 0.63	0.241		
TRE vs. Control groups	9	0.27	0.01 to 0.52	0.042	40.2%	*TRE: 76.6%* *HVRE: 66.3%*
HVRE vs. TRE	2*	-0.03	-0.47 to 0.41	0.892		
Five times sit-to-stand test						
HVRE vs. Control groups	3*	-0.65	-1.11 to -0.19	0.006		
TRE vs. Control groups	13*	-0.57	-0.85 to -0.29	<0.001	57.5%	*HVRE: 81.9%* *TRE: 68.0%*
HVRE vs. TRE	6*	-0.08	-0.52 to 0.36	0.720		
30-sec sit-to-stand test						
HVRE vs. Control groups	6*	1.08	0.66 to 1.49	<0.001		
TRE vs. Control groups	18*	0.96	0.67 to 1.25	<0.001	67.7%	*HVRE: 85.8%* *TRE: 64.2%*
HVRE vs. TRE	6*	0.12	-0.29 to 0.53	0.567		
6-min walking test						
HVRE vs. Control groups	3*	0.42	-0.20 to 1.02	0.183		
TRE vs. Control groups	13*	0.62	0.27 to 0.97	<0.001	68.8%	*TRE: 87.8%* *HVRE: 57.6%*
HVRE vs. TRE	4*	-0.21	-0.79 to 0.38	0.487		

*, Number of studies after omitting outliers; 95% CI, 95% confidence interval; HVRE, high-velocity resistance exercise; k, Number of comparisons; TRE, traditional resistance exercise; SMD, standardised mean difference.

eTable 12. Pairwise meta-analysis on leg press muscle power expressed in W.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1 n	Group 1 MD ± SD	Group 2 n	Group 2 MD ± SD	Between-group SMD (95% CI)
De Vos et al., 2005 *	PH	39.0%	69.0	26.0	High risk	84	68.33 ± 65.39	28	11.00 ± 46.00	0.93 (95% CI: 0.49 to 1.38)
Marsh et al., 2009 *	PI	31.1%	74.8	30.4	High risk	15	137.64 ± 318.42	15	25.93 ± 267.78	0.37 (95% CI: -0.35 to 1.09)
Reid et al., 2008	PI	45.6%	74.2	29.0	High risk	23	61.33 ± 92.75	12	23.05 ± 66.96	0.44 (95% CI: -0.27 to 1.15)
Sayers & Gibson, 2012	PH and PI	30.6%	70.6	28.8	High risk	25	198.50 ± 229.85	22	6.90 ± 157.58	0.94 (95% CI: 0.34 to 1.55)
Overall effect	***High-velocity resistance exercise vs. control group; I^2= 0%, τ^2= 0.00***									***0.76 (95% CI: 0.47 to 1.05)***
De Vreede et al., 2005	PH	0.0%	73.9	NR	Some concerns	34	10.80 ± 25.80	31	-7.00 ± 26.10	0.68 (95% CI: 0.18 to 1.18)
Englund et al., 2019	PI	40.0%	78.9	28.2	Some concerns	11	34.91 ± 96.03	13	15.70 ± 54.86	0.24 (95% CI: -0.56 to 1.05)
Marsh et al., 2009 *	PI	31.1%	74.8	30.4	High risk	15	16.74 ± 334.48	15	25.93 ± 267.78	-0.03 (95% CI: -0.75 to 0.69)
Reid et al., 2008	PI	45.6%	74.2	29.0	High risk	22	49.07 ± 59.29	12	23.05 ± 66.96	0.41 (95% CI: -0.30 to 1.12)
Sayers & Gibson, 2012	PH and PI	30.6%	70.6	28.8	High risk	25	138.70 ± 144.52	22	6.90 ± 157.58	0.86 (95% CI: 0.26 to 1.46)
Overall effect	***Traditional resistance exercise vs. control group, I^2= 9%, τ^2= 0.01***									***0.50 (95% CI: 0.20 to 0.80)***
Balachandran et al., 2014 *	PI	5.9%	71.3	32.6	Some concerns	8	231.00 ± 164.56	9	88.00 ± 149.70	0.87 (95% CI: -0.14 to 1.87)
Bottaro et al., 2007 *	PH	100.0%	66.5	21.2	High risk	11	190.18 ± 153.08	9	44.89 ± 115.38	1.01 (95% CI: 0.06 to 1.96)

Fielding et al., 2002 *	PI	0.0%	73.0	30.1	High risk	15	306.00 ± 177.78	15	32.00 ± 40.80	2.07 (95% CI: 1.16 to 2.98)
Marsh et al., 2009 *	PI	31.1%	74.8	30.4	High risk	15	137.64 ± 318.42	15	16.74 ± 334.48	0.36 (95% CI: -0.36 to 1.08)
Pamukoff et al., 2014	PH	55.0%	70.8	30.5	High risk	10	96.60 ± 126.49	10	95.90 ± 135.45	0.01 (95% CI: -0.87 to 0.88)
Reid et al., 2008	PI	45.6%	74.2	29.0	High risk	23	61.33 ± 92.75	22	49.07 ± 59.29	0.15 (95% CI: -0.43 to 0.74)
Sayers & Gibson, 2012	PH and PI	30.6%	70.6	28.8	High risk	25	198.50 ± 229.85	25	138.70 ± 144.52	0.31 (95% CI: -0.25 to 0.86)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 63%, τ^2= 0.25***									***0.62 (95% CI: 0.14 to 1.10)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 13. Pairwise meta-analysis on leg press muscle power expressed in $W.kg^{-1}$.

Authors	Health status	Males, %	Age, yrs	BMI, $kg.m^{-2}$	Risk of bias	Group 1		Group 2		Between-group SMD
						n	MD ± SD	n	MD ± SD	(95% CI)
Caserotti et al., 2008 [60 yrs] *	PH	0.0%	62.7	NR	High risk	20	0.57 ± 1.21	20	0.12 ± 1.25	0.36 (95% CI: -0.27 to 0.98)
Caserotti et al., 2008 [80 yrs] *	PI	0.0%	81.8	NR	High risk	12	0.81 ± 0.89	13	0.09 ± 0.92	0.77 (95% CI: -0.05 to 1.59)
Reid et al., 2008	PI	45.6%	74.2	29.0	High risk	23	6.08 ± 6.23	12	0.86 ± 4.50	0.89 (95% CI: 0.16 to 1.62)
Overall effect	***High-velocity resistance exercise vs. control group; I^2= 0%, τ^2= 0.00***									***0.63 (95% CI: 0.22 to 1.04)***
Reid et al., 2008	PI	45.6%	74.2	29.0	High risk	22	2.86 ± 3.47	12	0.86 ± 4.50	0.51 (95% CI: -0.21 to 1.22)
Overall effect	***Insufficient number of studies for Traditional resistance exercise vs. control group***									*-*
Bean et al., 2009 *	PI	31.2%	78.0	26.3	Some concerns	72	1.05 ± 3.31	66	0.00 ± 3.29	0.32 (95% CI: -0.02 to 0.65)
Reid et al., 2008	PI	45.6%	74.2	29.0	High risk	23	6.08 ± 6.23	22	2.86 ± 3.47	0.62 (95% CI: 0.02 to 1.22)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 0%, τ^2= 0.00***									***0.39 (95% CI: 0.09 to 0.68)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 14. Pairwise meta-analysis on stair climbing muscle power.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1 n	Group 1 MD ± SD	Group 2 n	Group 2 MD ± SD	Between-group SMD (95% CI)
Henwood & Taaffe, 2006 *	PH	NR	70.0	26.6	High risk	23	23.50 ± 66.84	22	-2.60 ± 75.94	0.36 (95% CI: -0.23 to 0.95)
Overall effect	***Insufficient number of studies for high-velocity resistance exercise vs. control group***									***-***
Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	22	-1.30 ± 68.22	22	-2.60 ± 75.94	0.02 (95% CI: -0.57 to 0.61)
Raj et al., 2012 *	PH	0.0%	67.5	NR	High risk	12	12.00 ± 119.05	10	6.00 ± 100.01	0.05 (95% CI: -0.79 to 0.89)
Vezzoli et al., 2019 *	PI	45.7%	72.4	27.2	High risk	20	23.00 ± 134.40	15	6.30 ± 86.09	0.14 (95% CI: -0.53 to 0.81)
Overall effect	***Traditional resistance exercise vs. control group, I^2= 0%, τ^2= 0.00***									***0.07 (95% CI: -0.33 to 0.46)***
Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	23	23.50 ± 66.84	22	-1.30 ± 68.22	0.36 (95% CI: -0.23 to 0.95)
Overall effect	***Insufficient number of studies for high-velocity resistance exercise vs. Traditional resistance exercise***									***-***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 15. Pairwise meta-analysis on stair climbing muscle power.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1		Group 2		Between-group SMD (95% CI)
						n	MD ± SD	n	MD ± SD	
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	11.10 ± 11.95	9	0.90 ± 4.33	1.04 (95% CI: 0.09 to 2.00)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	7.50 ± 11.19	10	-2.60 ± 5.70	1.07 (95% CI: 0.15 to 2.00)
Overall effect	***Insufficient number of studies for high-velocity resistance exercise vs. control group***									**-**
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	8	14.60 ± 9.54	9	0.90 ± 4.33	1.80 (95% CI: 0.62 to 2.97)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	5.20 ± 12.46	10	-2.60 ± 5.70	0.76 (95% CI: -0.13 to 1.65)
Van Roie et al., 2020 *	PH	100.0%	69.4	26.6	High risk	12	37.90 ± 64.10	14	15.20 ± 64.00	0.34 (95% CI: -0.43 to 1.12)
Overall effect	***Traditional resistance exercise vs. control group, I^2= 51%, τ^2= 0.24***									***0.87 (95% CI: 0.10 to 1.64)***
Coelho-Junior et al., 2021 [frail] *	PI	35.7%	76.0	24.8	High risk	11	11.10 ± 11.95	8	14.60 ± 9.54	-0.30 (95% CI: -1.22 to 0.61)
Coelho-Junior et al., 2021 [prefrail] *	PI	6.3%	65.0	26.3	High risk	11	7.50 ± 11.19	11	5.20 ± 12.46	0.19 (95% CI: -0.65 to 1.02)
Jaque et al., 2020 *	PH	0.0%	71.6	28.5	Some concerns	14	142.60 ± 119.41	12	34.70 ± 128.57	0.84 (95% CI: 0.03 to 1.66)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 42%, τ^2= 0.14***									***0.27 (95% CI: -0.38 to 0.92)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 16. Pairwise meta-analysis on countermovement jump muscle power.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1 n	Group 1 MD ± SD	Group 2 n	Group 2 MD ± SD	Between-group SMD (95% CI)
Caserotti et al., 2008 [60 yrs] *	PH	0.0%	62.7	NR	High risk	20	1.30 ± 2.91	20	-1.20 ± 4.51	0.65 (95% CI: 0.01 to 1.28)
Caserotti et al., 2008 [80 yrs] *	PI	0.0%	81.8	NR	High risk	12	1.10 ± 3.96	13	-0.60 ± 2.17	0.52 (95% CI: -0.28 to 1.32)
Valls et al., 2014 *	PH	52.2%	72	23.9	High risk	13	4.40 ± 6.75	10	-0.50 ± 6.32	0.72 (95% CI: -0.14 to 1.57)
Overall effect	***High-velocity resistance exercise vs. control group; I^2= 0%, τ^2= 0.00***									***0.63 (95% CI: 0.20 to 1.06)***
Raj et al., 2012 *	PH	0.0%	67.5	NR	High risk	12	70.00 ± 483.35	10	-43.00 ± 388.91	0.25 (95% CI: -0.60 to 1.09)
Overall effect	***Insufficient number of studies for Traditional resistance exercise vs. control group***									***-***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 17. Pairwise meta-analysis on leg press muscle strength.

Authors	Health status	Males, %	Age, yrs	BMI, kg.m^{-2}	Risk of bias	Group 1 n	Group 1 MD ± SD	Group 2 n	Group 2 MD ± SD	Between-group SMD (95% CI)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	46.11 ± 17.24	19	-2.21 ± 14.00	3.02 (95% CI: 2.05 to 3.99)
Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	23	7.40 ± 20.57	22	2.10 ± 28.30	0.21 (95% CI: -0.38 to 0.80)
Lopes et al., 2014 *	PH	0.0%	65.3	NR	High risk	10	7.66 ± 14.93	11	0.18 ± 14.15	0.49 (95% CI: -0.38 to 1.37)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	14.67 ± 3.92	11	4.14 ± 4.82	2.32 (95% CI: 1.22 to 3.42)
Miszko et al., 2003 *	PH	43.6%	72.5	NR	High risk	11	13.20 ± 32.71	15	4.10 ± 38.22	0.24 (95% CI: -0.54 to 1.03)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	26.00 ± 17.76	15	0.00 ± 21.88	1.27 (95% CI: 0.48 to 2.06)
Ramírez-Campillo et al., 2017 *	PH	0.0%	66.6	30.8	Some concerns	58	14.81 ± 22.18	15	0.00 ± 21.88	0.66 (95% CI: 0.08 to 1.24)
Wallerstein et al., 2012 *	PH	NR	63.8	NR	High risk	20	32.93 ± 44.10	19	-5.90 ± 34.57	0.96 (95% CI: 0.29 to 1.62)
Overall effect					***High-velocity resistance exercise vs. control group; I^2= 80%, r^2= 0.59***					***1.08 (95% CI: 0.48 to 1.68)***
Cavalcante et al., 2020 *	PH	26.7%	71.0	28.6	High risk	23	28.00 ± 15.13	22	-3.00 ± 28.16	1.36 (95% CI: 0.70 to 2.01)
Charette et al., 1991 *	PH	0.0%	69.1	NR	High risk	13	16.80 ± 11.57	6	4.90 ± 11.10	0.99 (95% CI: -0.04 to 2.03)
Fatouros et al., 2002 *	PH	100.0%	70.4	29.3	High risk	8	50.60 ± 8.74	8	1.70 ± 8.45	5.38 (95% CI: 3.02 to 7.74)
Fernández-Lezaun et al., 2017 [Men] *	PH	100.0%	70.2	28.2	High risk	29	19.71 ± 27.24	11	-2.62 ± 20.13	0.86 (95% CI: 0.13 to 1.58)

Fernández-Lezaun et al., 2017 [Women] *	PH	0.0%	69	27.1	High risk	39	20.26 ± 20.44	9	-1.06 ± 20.87	1.02 (95% CI: 0.27 to 1.78)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	17	41.17 ± 18.47	19	-2.21 ± 14.00	2.61 (95% CI: 1.69 to 3.52)
Griffen et al., 2022	PH	0.0%	67.0	25.1	High risk	9	29.71 ± 5.76	9	2.84 ± 3.80	5.25 (95% CI: 3.10 to 7.39)
Haykowsky et al., 2000 *	PH	100.0%	68	NR	High risk	12	82.00 ± 47.51	10	-1.00 ± 56.24	1.55 (95% CI: 0.57 to 2.53)
Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	22	7.50 ± 18.99	22	2.10 ± 28.30	0.22 (95% CI: -0.37 to 0.81)
Lopes et al., 2014 *	PH	0.0%	65.3	NR	High risk	13	11.31 ± 23.94	11	0.18 ± 14.15	0.53 (95% CI: -0.28 to 1.35)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	14	15.41 ± 4.17	11	4.14 ± 4.82	2.44 (95% CI: 1.36 to 3.52)
Mafi et al., 2019	PI	100.0%	68.5	23.4	Some concerns	14	7.92 ± 13.75	16	-1.20 ± 0.22	0.95 (95% CI: 0.19 to 1.71)
McCartney et al., 1995 [60-70 Men] *	PH	100.0%	63.6	NR	High risk	22	41.56 ± 32.48	16	-6.06 ± 31.61	1.45 (95% CI: 0.72 to 2.18)
McCartney et al., 1995 [60-70 Women] *	PH	0.0%	65	NR	High risk	17	22.51 ± 15.89	19	3.46 ± 20.14	1.02 (95% CI: 0.32 to 1.72)
McCartney et al., 1995 [70-80 Men] *	PH	100.0%	72.7	NR	High risk	17	31.17 ± 34.21	8	-0.87 ± 25.64	0.97 (95% CI: 0.08 to 1.86)
McCartney et al., 1995 [70-80 Women] *	PH	0.0%	72.1	NR	High risk	20	23.38 ± 22.71	23	7.80 ± 15.87	0.79 (95% CI: 0.17 to 1.42)
Miszko et al., 2003 *	PH	43.6%	72.5	NR	High risk	13	19.66 ± 49.62	15	4.10 ± 38.22	0.34 (95% CI: -0.40 to 1.09)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	21.30 ± 23.00	15	0.00 ± 21.88	0.92 (95% CI: 0.17 to 1.68)
Rhodes et al., 2000 *	PH	0.0%	68.8	NR	High risk	22	19.00 ± 9.78	22	0.60 ± 9.08	1.91 (95% CI: 1.19 to 2.64)

Taaffe et al., 1999 *	PH	64.2%	NR	24.4	High risk	34	29.92 ± 25.40	12	-0.40 ± 28.61	1.14 (95% CI: 0.43 to 1.84)
Vale et al., 2017 *	PH	0.0%	67.5	28.6	High risk	10	37.50 ± 12.50	10	4.50 ± 25.85	1.56 (95% CI: 0.53 to 2.59)
Van Roie et al., 2020 *	PH	100.0%	69.4	26.6	High risk	12	50.10 ± 51.46	14	-0.40 ± 51.85	0.95 (95% CI: 0.13 to 1.77)
Wallerstein et al., 2012 *	PH	NR	63.8	NR	High risk	20	47.17 ± 54.31	19	-5.90 ± 34.57	1.14 (95% CI: 0.45 to 1.82)
Overall effect					***Traditional resistance exercise vs. control group, I^2= 67%, τ^2= 0.34***					***1.26 (95% CI: 0.96 to 1.56)***
Balachandran et al., 2014 *	PI	5.9%	71.3	32.6	High risk	8	14.60 ± 42.93	9	23.60 ± 35.30	-0.22 (95% CI: -1.17 to 0.74)
Bottaro et al., 2007 *	PH	100.0%	66.5	21.2	High risk	11	47.27 ± 38.48	9	47.22 ± 33.47	0.00 (95% CI: -0.88 to 0.88)
Filho et al., 2022 *	PH	0.0%	67.0	27.5	High risk	18	46.11 ± 17.24	17	41.17 ± 18.47	0.27 (95% CI: -0.40 to 0.94)
Henwood et al., 2006 *	PH	NR	70.0	26.6	High risk	23	7.40 ± 20.57	22	7.50 ± 18.99	0.00 (95% CI: -0.59 to 0.58)
Lopes et al., 2014 *	PH	0.0%	65.3	NR	High risk	10	7.66 ± 14.93	13	11.31 ± 23.94	-0.17 (95% CI: -1.00 to 0.66)
Lopes et al., 2016 *	PH	0.0%	67.2	NR	High risk	12	14.67 ± 3.92	14	15.41 ± 4.17	-0.18 (95% CI: -0.95 to 0.60)
Miszko et al., 2003 *	PH	43.6%	72.5	NR	High risk	11	13.20 ± 32.71	13	19.66 ± 49.62	-0.15 (95% CI: -0.95 to 0.66)
Pamukoff et al., 2014	PH	55.0%	70.8	30.5	High risk	10	33.90 ± 49.29	10	22.40 ± 52.52	0.22 (95% CI: -0.66 to 1.10)
Ramírez-Campillo et al., 2014 *	PH	0.0%	67.2	30.7	High risk	15	26.00 ± 17.76	15	21.30 ± 23.00	0.22 (95% CI: -0.50 to 0.94)

Tiggemann et al., 2016 *	PH	0.0%	65.6	26.8	Some concerns	13	34.68 ± 14.34	12	34.68 ± 15.89	0.00 (95% CI: -0.78 to 0.78)
Wallerstein et al., 2012 *	PH	NR	63.8	NR	High risk	20	32.93 ± 44.10	20	47.17 ± 54.31	-0.28 (95% CI: -0.91 to 0.34)
Overall effect	***High-velocity resistance exercise vs. Traditional resistance exercise, I^2= 0%, τ^2= 0.00***									***-0.02 (95% CI: -0.25 to 0.20)***

*, Within-group mean difference estimated assuming a correlation of r = 0.05 between baseline and post-intervention assessment measures; 95% CI, 95% confidence interval; I^2, between-study variability; τ^2, between-study variance; MD, within-group mean difference; PH, physically healthy participants; PI, physically impaired participants; SMD, standardised mean difference.

eTable 18. Side-split analyses from network meta-analysis on leg press muscle power and muscle strength outcomes.

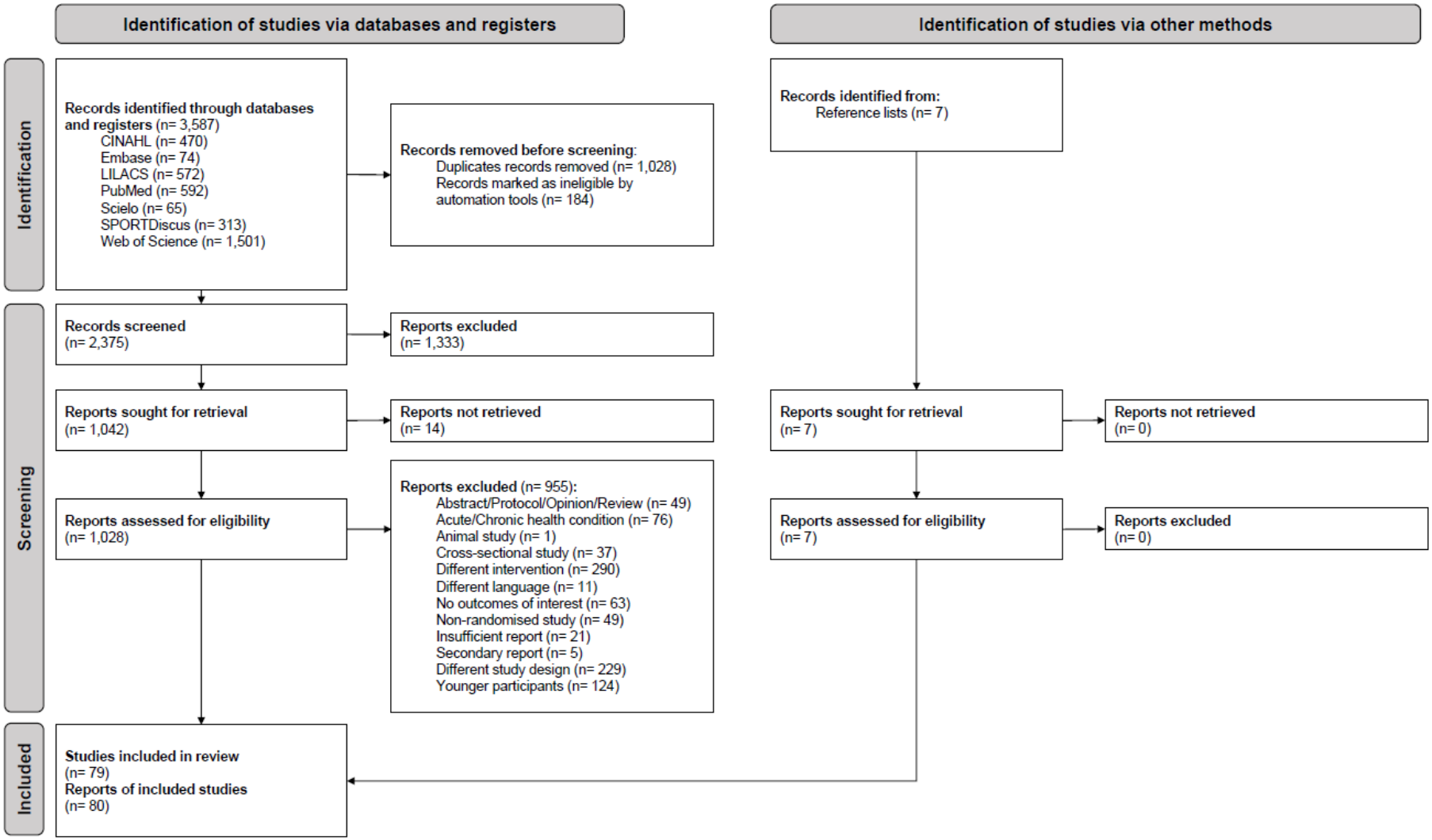
Comparisons	Direct comparison SMD (95% CI)	Indirect comparison SMD (95% CI)	Difference in estimates SMD (95% CI)	P-value
Leg press muscle power				
HVRE vs. Control groups	0.71 (0.25 to 1.16)	1.58 (0.71 to 2.46)	-0.87 (-1.86 to 0.11)	0.081
TRE vs. Control groups	0.47 (0.05 to 0.89)	-0.31 (-1.28 to 0.67)	0.77 (-0.29 to 1.83)	0.154
HVRE vs. TRE	0.59 (0.20 to 0.98)	0.26 (-0.77 to 1.29)	0.34 (-0.76 to 1.44)	0.548
Leg press muscle strength				
HVRE vs. Control groups	1.03 (0.60 to 1.45)	1.27 (0.64 to 1.91)	-0.25 (-1.01 to 0.52)	0.530
TRE vs. Control groups	1.23 (0.97 to 1.49)	0.72 (-0.45 to 1.88)	0.52 (-0.68 to 1.71)	0.397
HVRE vs. TRE	-0.02 (-0.39 to 0.34)	-0.54 (-1.38 to 0.30)	0.52 (-0.40 to 1.43)	0.269

95% CI, 95% confidence interval; HVRE, high-velocity resistance exercise; TRE, traditional resistance exercise; SMD, standardised mean difference.

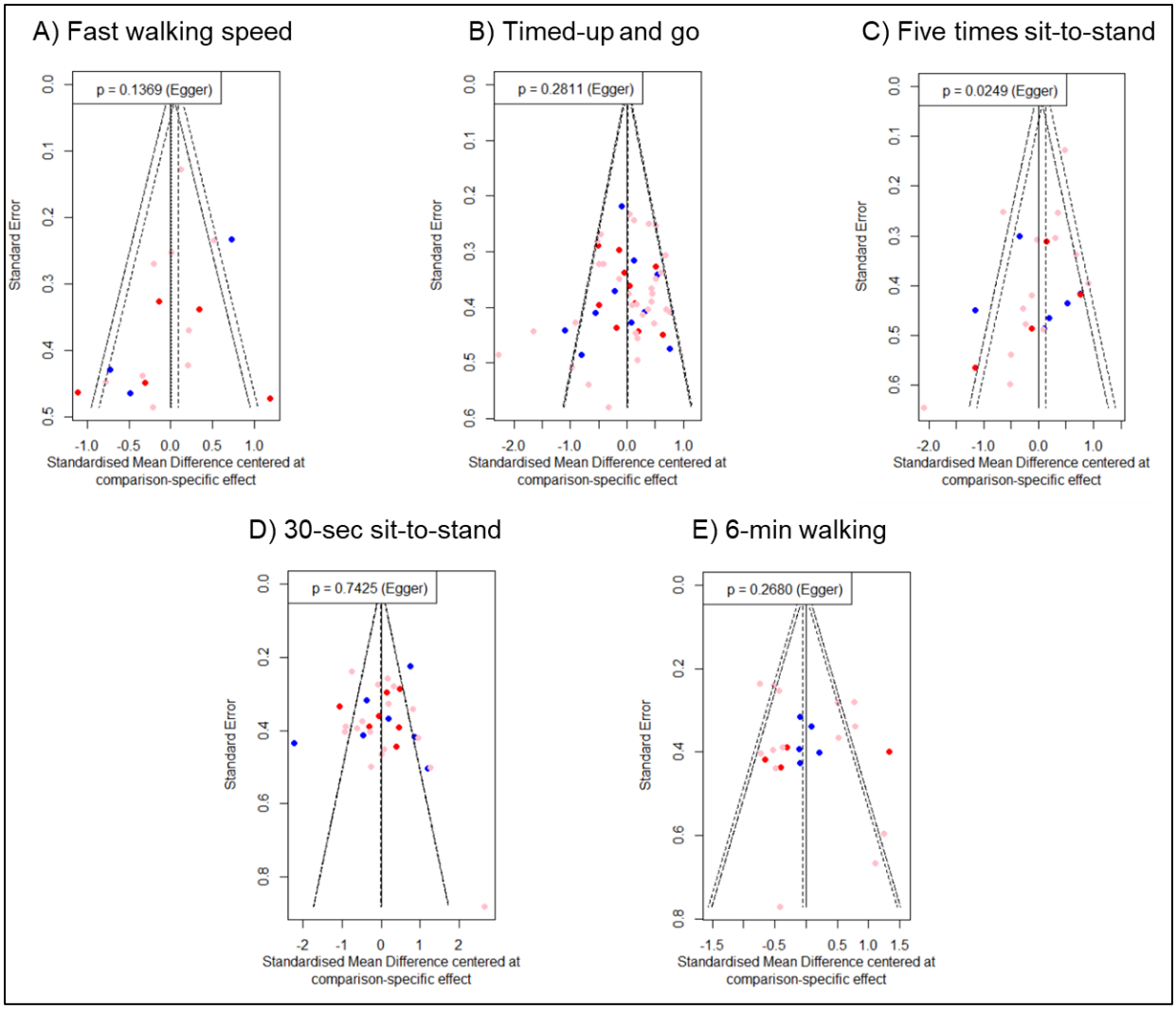
eTable 19. Sensitivity analyses by omitting outliers from network meta-analysis on muscle strength and muscle power outcomes.

Comparisons	k	ES	95% CI	P-value	I^2	Best intervention probability
Leg press power						
HVRE vs. Control groups	4	0.83	0.56 to 1.09	<0.001	0%	*HVRE: 99.7%* *TRE: 50.3%*
TRT vs. Control groups	5	0.48	0.22 to 0.75	<0.001		
HVRE vs. TRT	6*	0.34	0.08 to 0.61	0.012		
Leg press 1-RM						
HVRE vs. Control groups	7*	0.96	0.62 to 1.30	<0.001	50.7%	*TRE: 93.0%* *HVRE: 57.0%*
TRT vs. Control groups	22*	1.13	0.89 to 1.37	<0.001		
HVRE vs. TRT	10*	-0.18	-0.50 to 0.14	0.281		

*, Number of studies after omitting outliers; 95% CI, 95% confidence interval; HVRE, high-velocity resistance exercise; k, Number of comparisons; TRT, traditional resistance exercise; SMD, standardised mean difference.



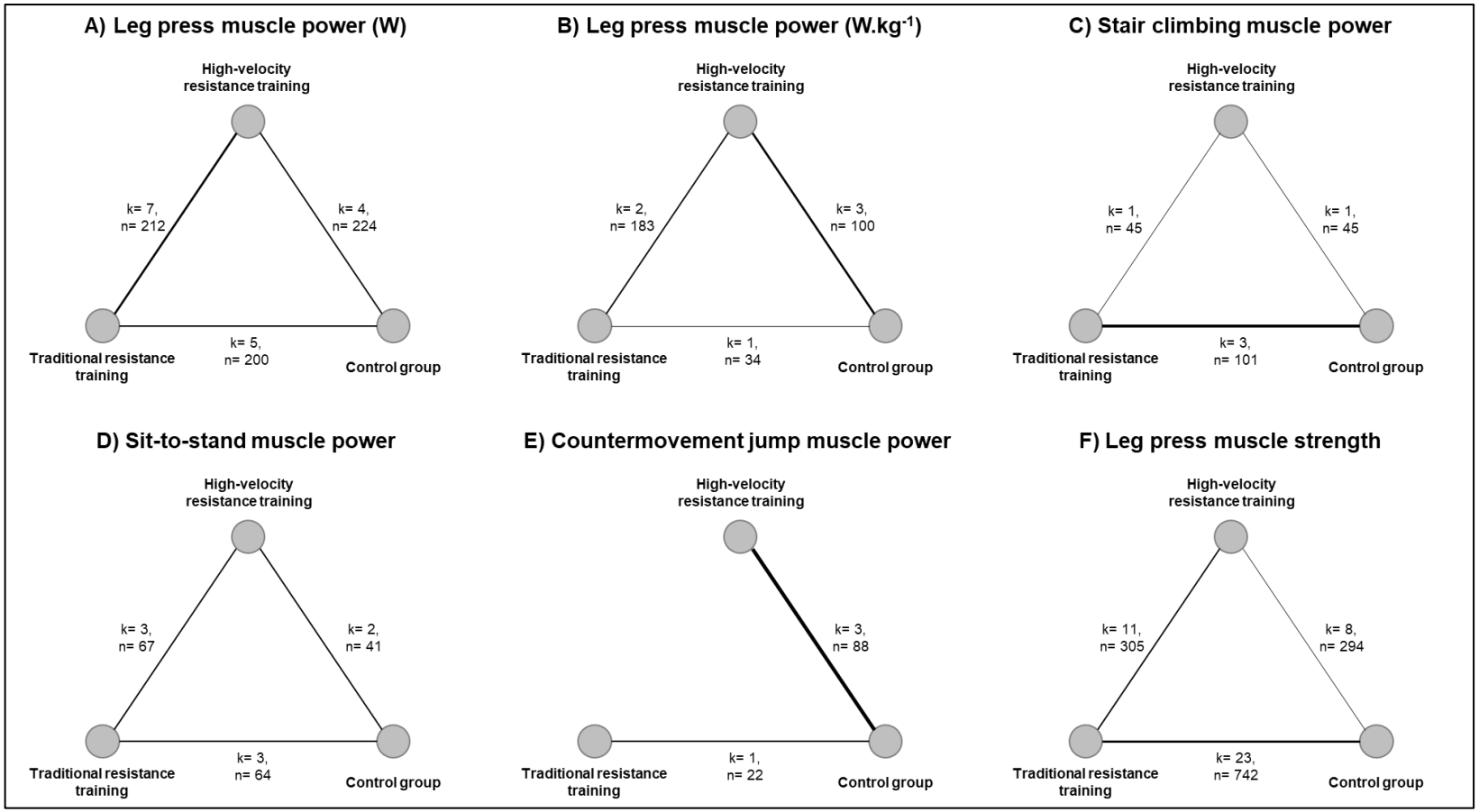
eFigure 1. Flow chart of study selection process.



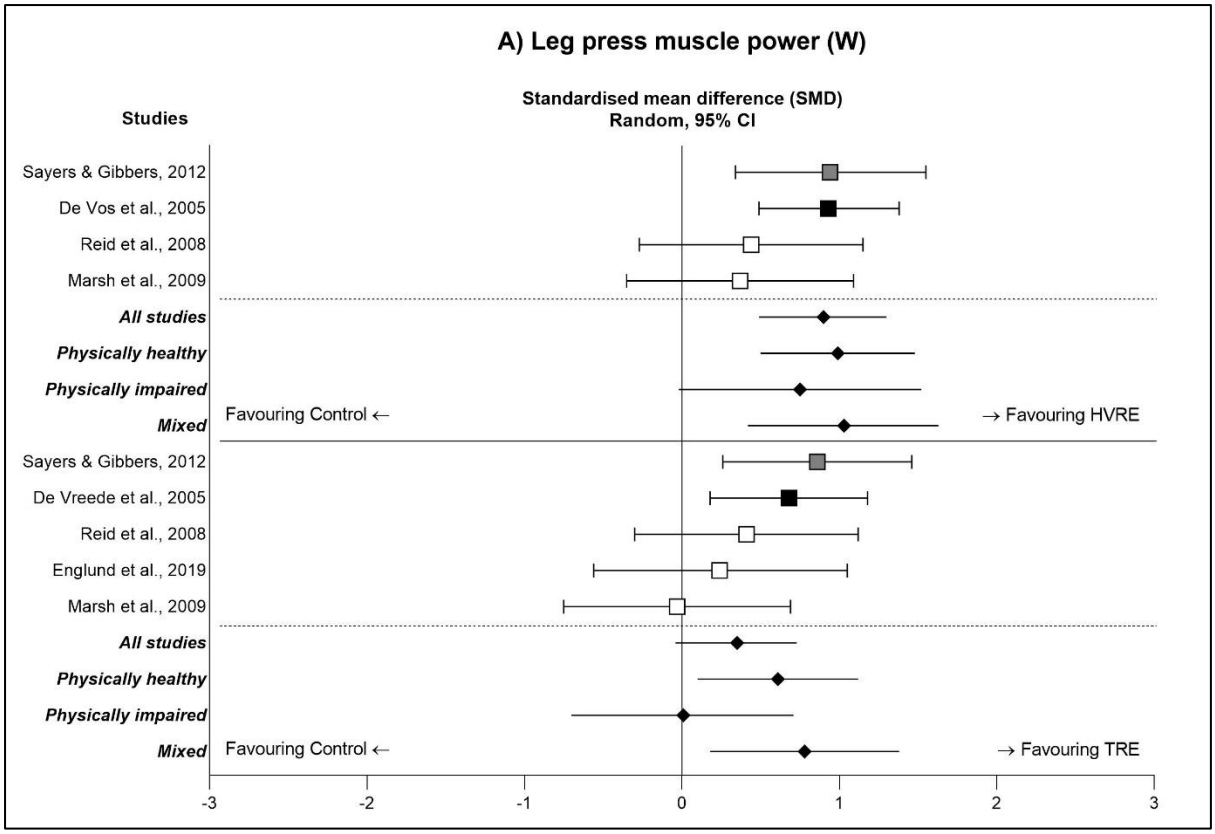
eFigure 2. Comparison-adjusted funnel plots for studies comparing high-velocity and low-to-moderate-velocity resistance exercise on A) fast walking speed, B) timed-up and go, C) five times sit-to-stand, D) 30-sec sit-to-stand and E) 6-min walking test in older adults. Blue dots indicate high-velocity resistance exercise comparisons; red dots indicate traditional resistance exercise comparisons; pink dots indicate control group comparisons.



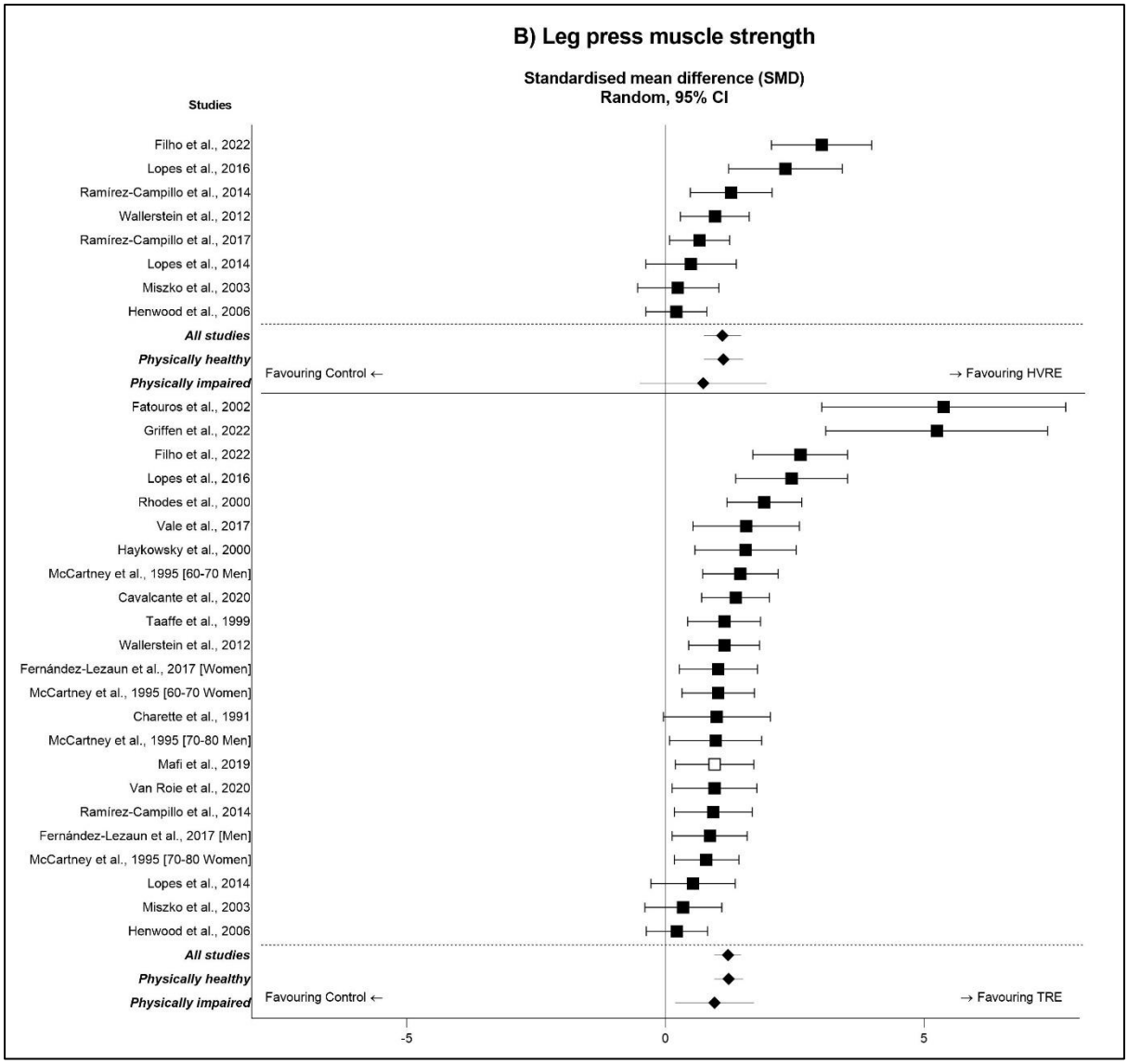
eFigure 3. Forward plots for Cook's distance for A) fast walking speed, B) five times sit-to-stand, C) 30-sec sit-to-stand and D) 6-min walking test.



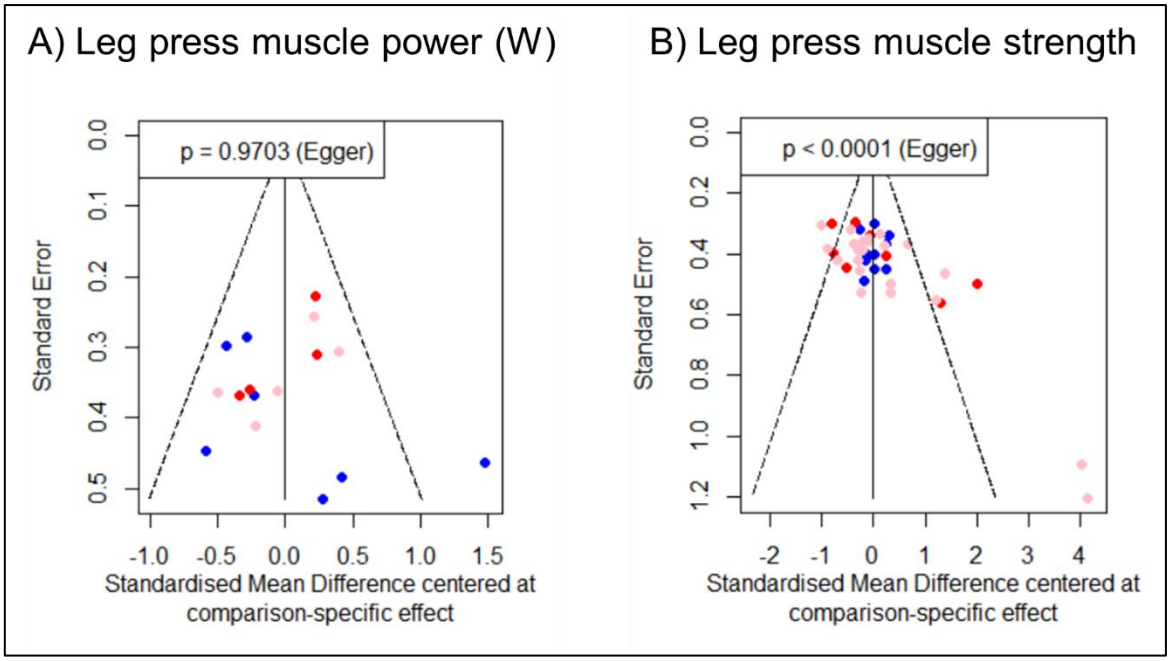
eFigure 4. Network geometry of studies examining A) leg press muscle power expressed in W, B) leg press muscle power expressed in W.kg^{-1}, C) stair climbing muscle power, D) sit-to-stand muscle power, E) countermovement jump muscle power and F) leg press muscle strength. k, number of comparisons; n, sample size.



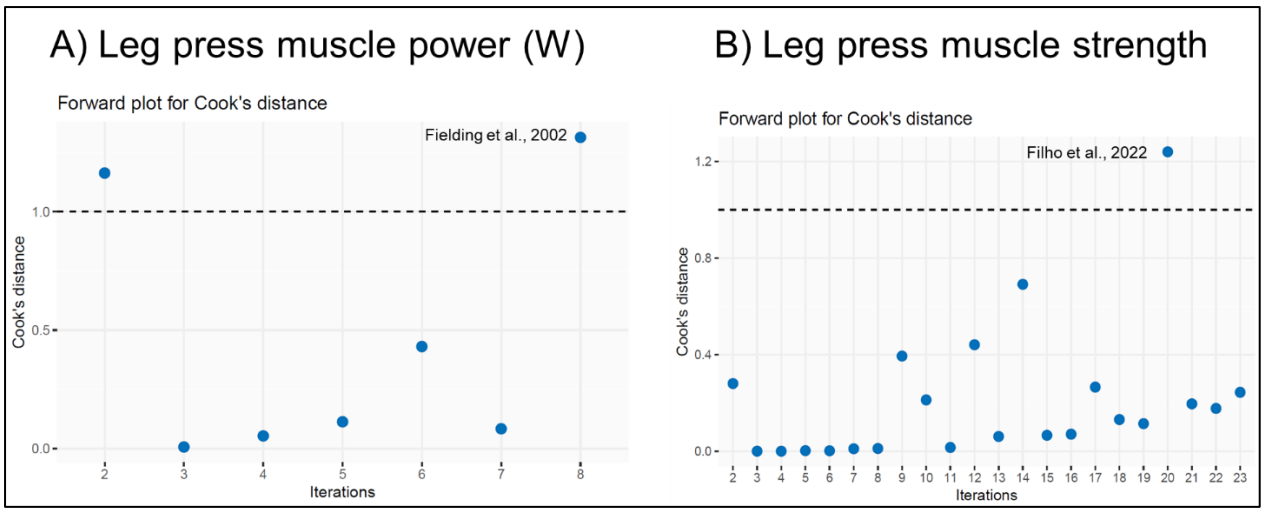
eFigure 5. Network meta-analysis on leg press muscle power (expressed in W) based on health status. White squares indicate studies involving physically impaired participants; grey squares indicate studies involving mixed participants; black squares indicate studies involving physical healthy participants. HVRE, high-velocity resistance exercise; TRE, traditional resistance exercise.



eFigure 6. Network meta-analysis on leg press muscle strength based on health status. White squares indicate studies involving physically impaired participants; black squares indicate studies involving physical healthy participants. HVRE, high-velocity resistance exercise; TRE, traditional resistance exercise.



eFigure 7. Comparison-adjusted funnel plots for studies comparing high-velocity and low-to-moderate-velocity resistance exercise on A) leg press muscle power (expressed in W) and B) leg press muscle strength. Blue dots indicate high-velocity resistance exercise comparisons; red dots indicate traditional resistance exercise comparisons; pink dots indicate control group comparisons.



eFigure 8. Forward plots for Cook's distance for A) leg press muscle power (expressed in W) and B) leg press muscle strength.